Theory of Computation

Languages and Finite Automata

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  Example: "cat", "dog", "house", ...
- Defined over an alphabet, Σ:
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- More Examples
  - **0011** and **11** are strings over $\Sigma = \{0, 1\}$
  - **abc** and **bbb** are strings over $\Sigma = \{a, b, \ldots, z\}$
  - **(()(()))** and **)(()** are strings over $\Sigma = \{(,)\}$

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Alphabets and Strings

- We will use small alphabets: $\Sigma = \{a, b\}$

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- **Strings**:

a	
ab	u = ab
abba	v = bbbaaa
baba	w = abba
aaabbbaabab	

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String Operations

- Consider two strings: w & v over $\Sigma = \{a, b\}$

$w = a_1 a_2 a_3 \ldots a_n$	abba
$v = b_1 b_2 b_3 \ldots b_n$	bbbaaa

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- **Concatenation**:

  $wv = a_1 a_2 a_3 \ldots a_n b_1 b_2 b_3 \ldots b_n$ | abbabbbaaa

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- **Reverse**:

  Consider the following string:

  $w = a_1 a_2 a_3 \ldots a_n$ | ababaaabbb

  After reverse:

  $w^R = a_n \ldots a_3 a_2 a_1$ | bbbaaababa

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  After reverse:

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- **String Length:** $w = a_1 a_2 a_3 \ldots a_n$

  Length: $|w| = n$

  Examples:

  $|abba| = 4$

  $|aa| = 2$

  $|a| = 1$

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String Operations

- **Recursive Definition of Length**

For any letter	$\|a\| = 1$
For any String	$\|wa\| = \|w\| + 1$

String Operations

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For any letter	$\lvert a \rvert = 1$
For any String	$\lvert wa \rvert = \lvert w \rvert + 1$
Example: $\lvert abba \rvert$	$= \lvert abb \rvert + 1$
	$= \lvert ab \rvert + 1 + 1$
	$= \lvert a \rvert + 1 + 1 + 1$
	$= 1 + 1 + 1 + 1$
	$= 4$

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- **Length of Concatenation:** $\lvert uv\rvert = \lvert u\rvert + \lvert v\rvert$

u = aab	$\lvert u\rvert = 3$
v = abaab	$\lvert v\rvert = 5$
$\lvert uv\rvert = \lvert aababaab\rvert$	$= 8$
	$\lvert uv\rvert = \lvert u\rvert + \lvert v\rvert = 3 + 5 = 8$

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String Operations

- **Proof of Concatenation Length**
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  From the definition of length: $|uv| = |u| + 1 = |u| + |v|$
  Inductive hypothesis: $|uv| = |u| + |v|$ for $|v| = 1, 2, 3, \ldots, n$
  Inductive step: we will prove $|uv| = |u| + |v|$ for $|v| = n + 1$
  Write $v = wa$, where $|w| = n$, $|a| = 1$
  from definition of length $|uv| = |uwa| = |uw| + 1$; $|wa| = |w| + 1$
  From inductive hypothesis: $|uw| = |u| + |w|$
  Thus, $|uv| = |u| + |w| + 1 = |u| + |wa| = |u| + |v|$

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String Operations

- **Empty String:** A string with no letters, λ
  Observations: $|\lambda| = 0$
  $\lambda w = w\lambda = w$
  $\lambda abba = abba\lambda = abba$

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- **Substring of string:** a subsequence of consecutive characters

String	**Substring**
abbab	ab
abbab	abba
abbab	b
abbab	bbab

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String	Substring
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abbab	b
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- **Prefix and Suffix:** Consider a String: *abbab*

Prefixes	Suffixes
λ	*abbab*
a	bbab
ab	bab
abb	ab
abba	b

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String Operations

- **Another Operation:**
  $w^n = \underbrace{www\ldots w}_{n}$
  Example: $(abba)^2 = abbaabba$
  Definition: $w^0 = \lambda$
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- **The * Operation:**
  Σ^* = the set of all possible strings from alphabet Σ
  $\Sigma = \{a, b\}$
  $\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \ldots\}$

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  Σ^* = the set of all possible strings from alphabet Σ
  $\Sigma = \{a, b\}$
  $\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \ldots\}$
- **The + Operation:**
  Σ^+ = the set of all possible strings from alphabet Σ except λ
  $\Sigma = \{a, b\}$
  $\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \ldots\}$
  $\Sigma^+ = \Sigma^* - \lambda$
  $\Sigma^+ = \{a, b, aa, ab, ba, bb, aaa, aab, \ldots\}$

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  - **Example:**
    $\Sigma = \{a, b\}$
    $\Sigma^* = \{\lambda, a, b, aa, ab, ba, bb, aaa, aab, \ldots\}$
  - **Languages:**
    $\{\lambda\}$
    $\{a, aa, aab\}$
    $\{\lambda, a, abba, baba, aa, ab, aaaaaa\}$

Language

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  - **Languages:**
    $\{\lambda\}$
    $\{a, aa, aab\}$
    $\{\lambda, a, abba, baba, aa, ab, aaaaaa\}$
  - An **infinite language** $L = \{a^n b^n : n \geq 0\}$

$$\left.\begin{array}{l} \lambda \\ ab \\ aabb \\ aaaaabbbbb \end{array}\right\} \in \mathsf{L} \qquad\qquad abb \notin L$$

Operations on Languages

- **The usual set operations:**
  $\{a, ab, aaaa\} \bigcup \{bb, ab\} = \{a, ab, bb, aaaa\}$
  $\{a, ab, aaaa\} \bigcap \{bb, ab\} = \{ab\}$
  $\{a, ab, aaaa\} - \{bb, ab\} = \{a, aaaa\}$

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- **Complement:** $\overline{L} = \Sigma^* - L$
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- **Reverse:** $L^R = \{w^R : w \in L\}$
  $\{ab, aab, baba\}^R = \{ba, baa, abab\}$
  $L = \{a^n b^n : n \geq 0\}$
  $L^R = \{b^n a^n : n \geq 0\}$

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  $\{ab, aab, baba\}^R = \{ba, baa, abab\}$
  $L = \{a^n b^n : n \geq 0\}$
  $L^R = \{b^n a^n : n \geq 0\}$
- **Concatenation:** $L_1 L_2 = \{xy : x \in L_1, y \in L_2\}$
  $\{a, ab, ba\}\{b, aa\} = \{ab, aaa, abb, abaa, bab, baaa\}$

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  $\{ab, aab, baba\}^R = \{ba, baa, abab\}$
  $L = \{a^n b^n : n \geq 0\}$
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- **Concatenation:** $L_1 L_2 = \{xy : x \in L_1, y \in L_2\}$
  $\{a, ab, ba\}\{b, aa\} = \{ab, aaa, abb, abaa, bab, baaa\}$
- **Another Operation:** $L^n = \underbrace{LL \ldots L}_{n}$
  $\{a, b\}^3 = \{a, b\}\{a, b\}\{a, b\} =$
  $\{aaa, aab, aba, abb, baa, bab, bba, bbb\}$
  $L^0 = \{\lambda\}$; $\{a, bb, aa\}^0 = \{\lambda\}$

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Operations on Languages

- **Star-Closure (Kleene*) :** $\boxed{L^* = L^0 \bigcup L^1 \bigcup L^2 \ldots}$

  Example:

$$\{a, bb\}^* = \left\{ \begin{array}{c} \lambda, \\ a, bb, \\ aa, abb, bba, bbbb, \\ aaa, aabb, abba, abbbb, \ldots \end{array} \right\}$$

- **Positive Closure:** $\boxed{L^+ = L^1 \bigcup L^2 \bigcup L^3 \ldots = L^* - \{\lambda\}}$

  Example:

$$\{a, bb\}^+ = \left\{ \begin{array}{c} a, bb, \\ aa, abb, bba, bbbb, \\ aaa, aabb, abba, abbbb, \ldots \end{array} \right\}$$

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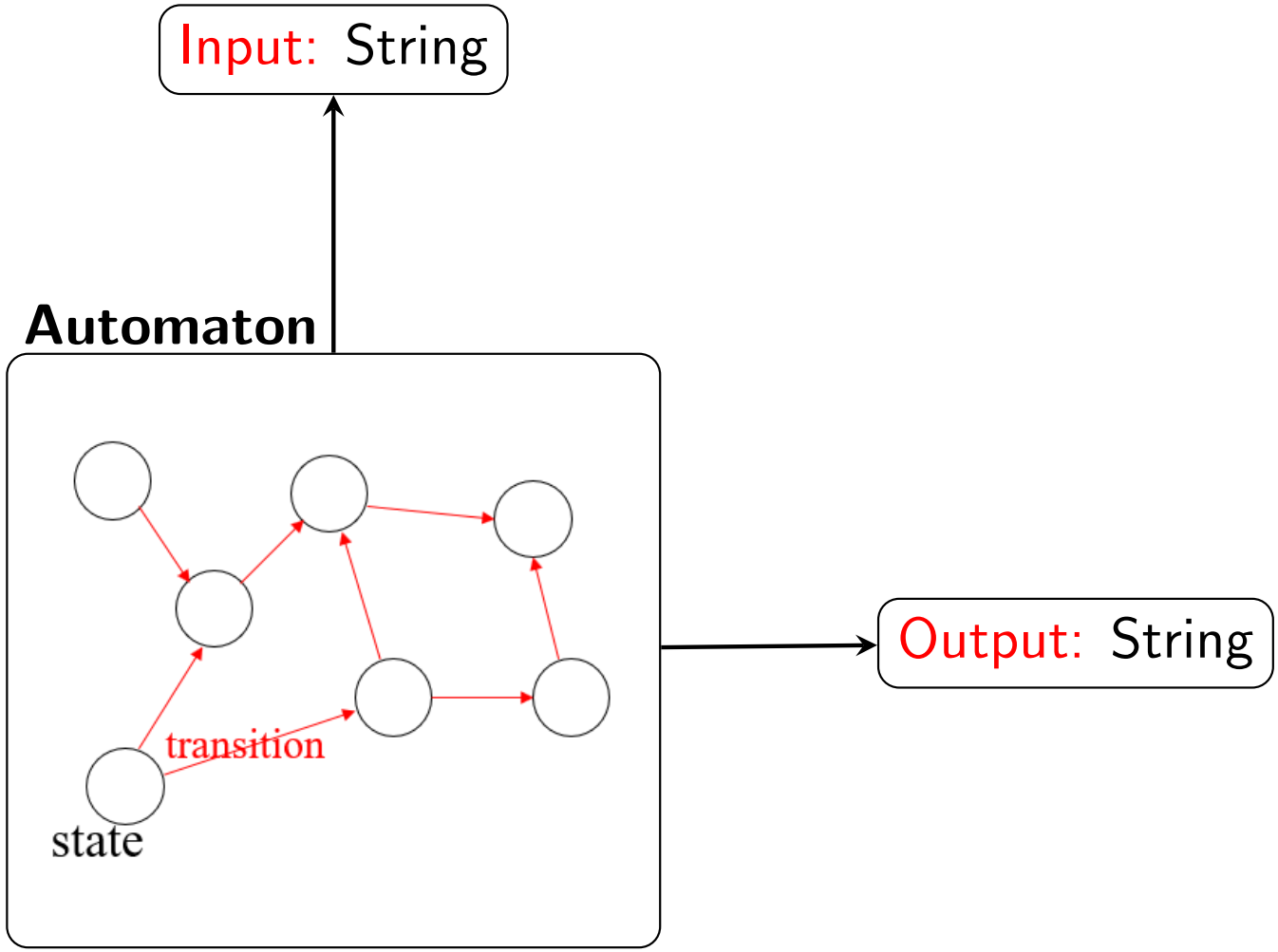
Finite Automata



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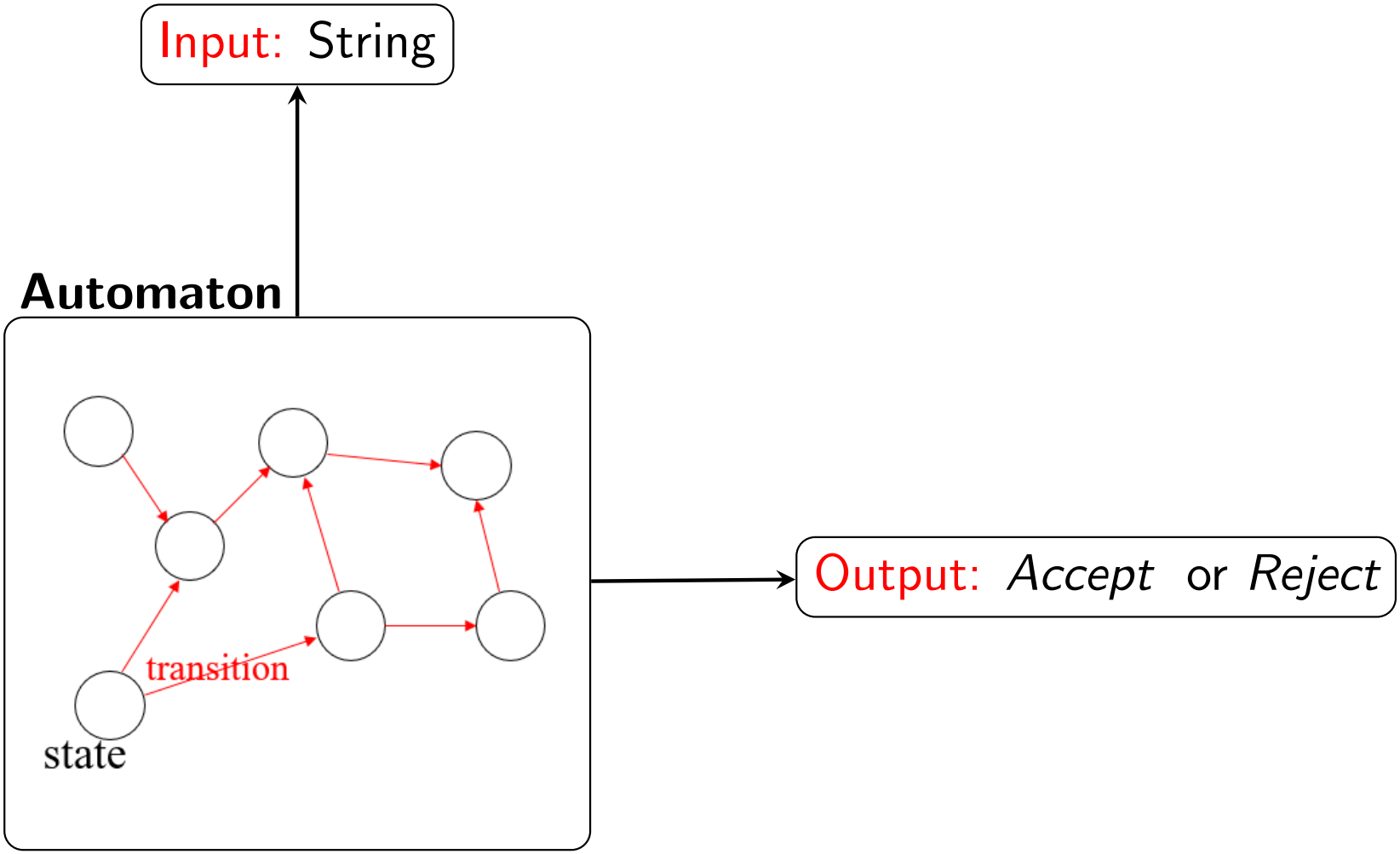
Finite Acceptor



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Finite Automata or Finite State Machines

- Problem is to decide whether a given string is a member of some paricular language or not.

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Types of Finite State Machines

- Finite Automata (Accept / Reject):
  - Deterministic Finite Automata (DFA)
  - Non-deterministic Finite Automata (NFA)
  - Finite Automata with ϵ-transitions (ϵ-NFA)
- Finite Automata (Generate an output sequence):
  - Mealy And Moore Machine

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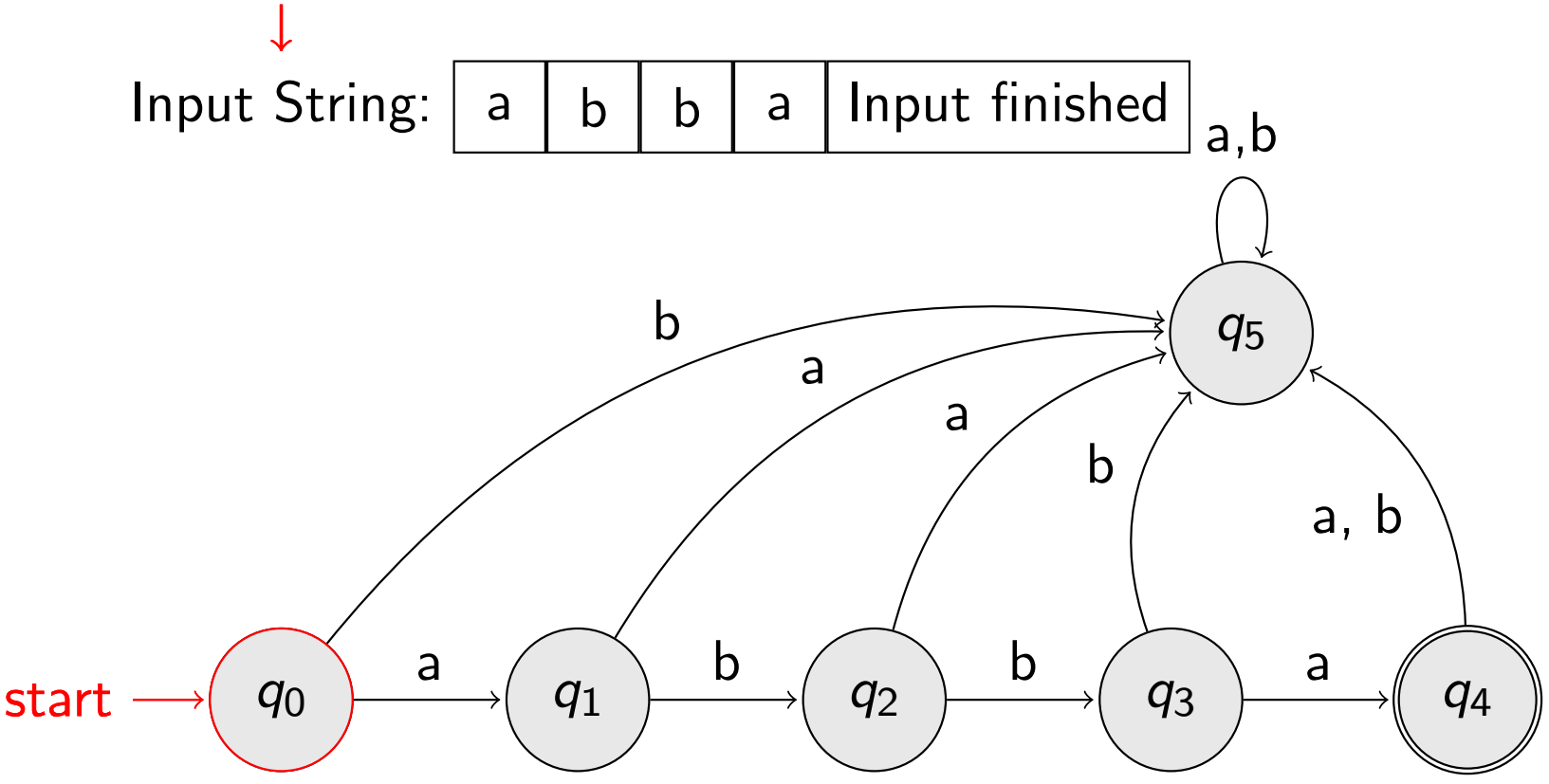
Transition Graph

- **abba** -Finite Acceptor:



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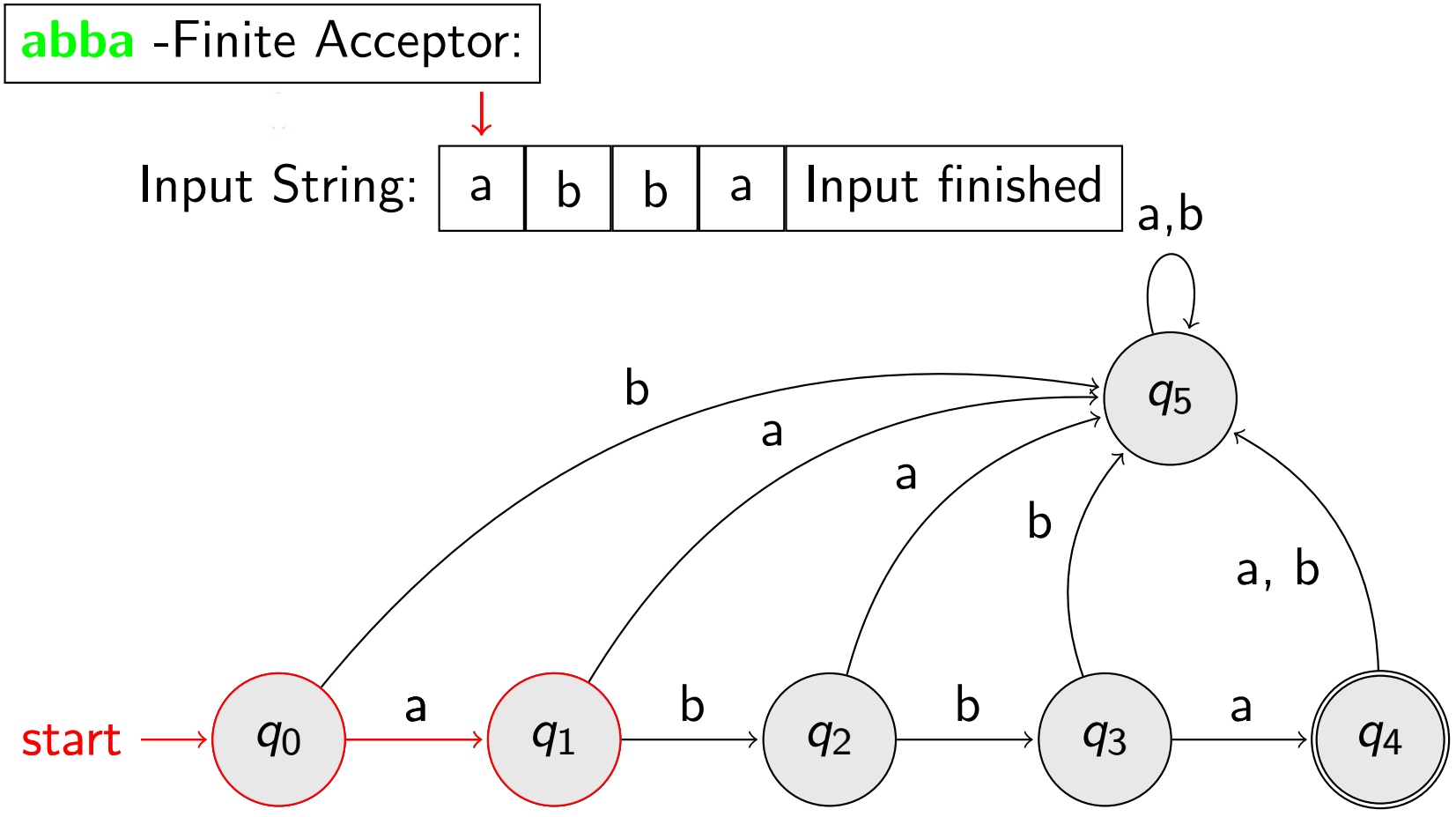

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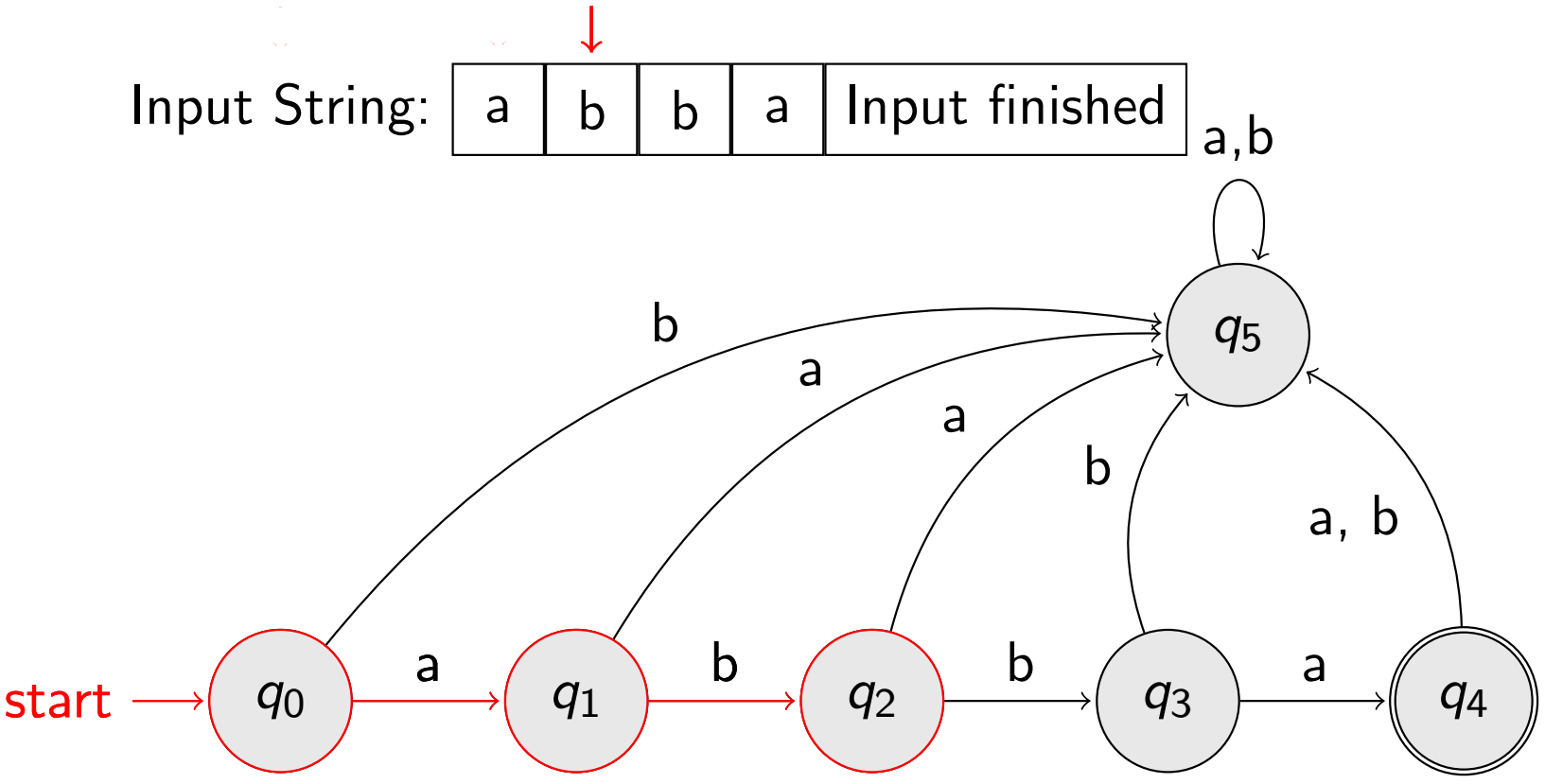

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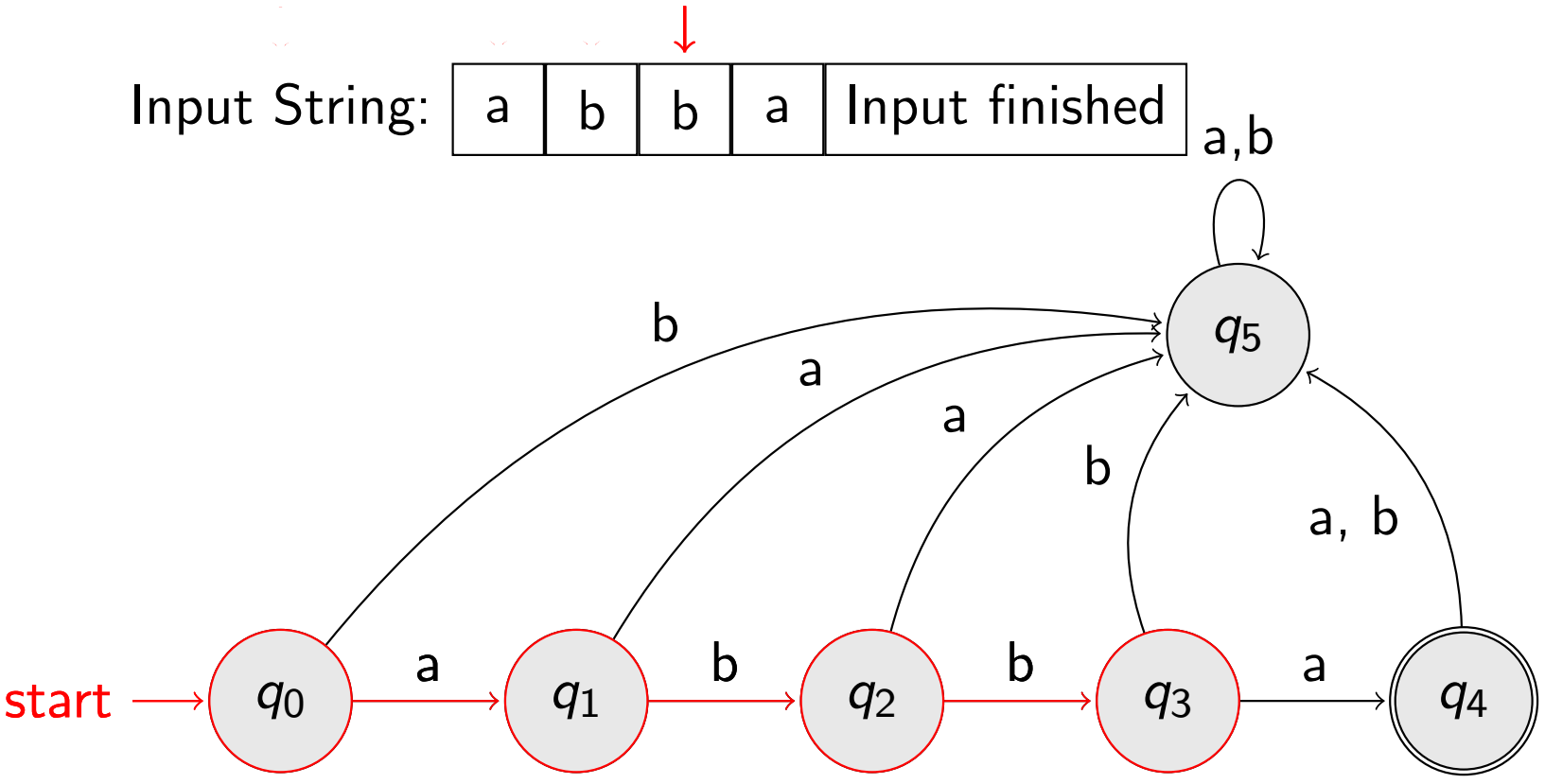

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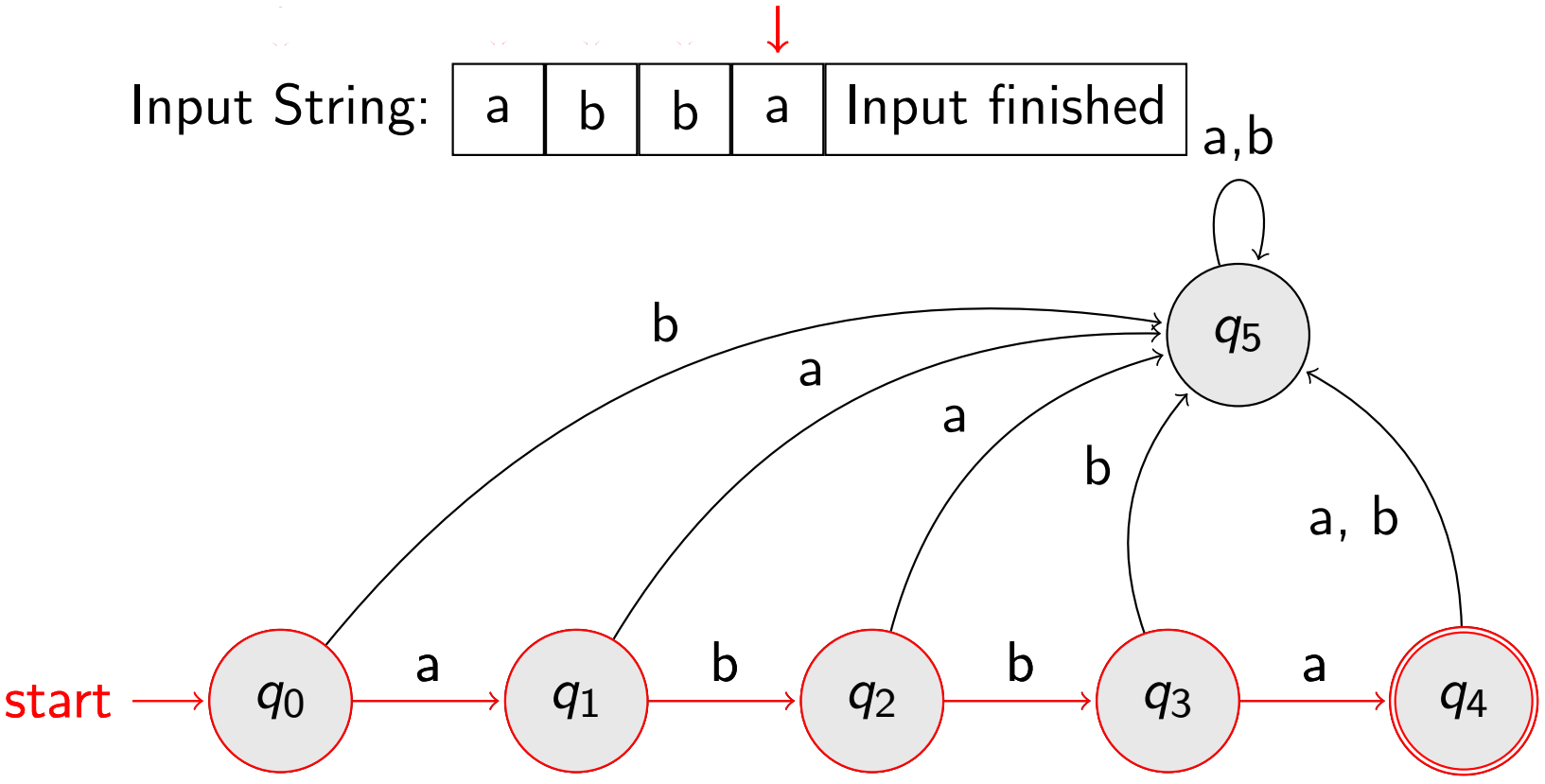

Transition Graph

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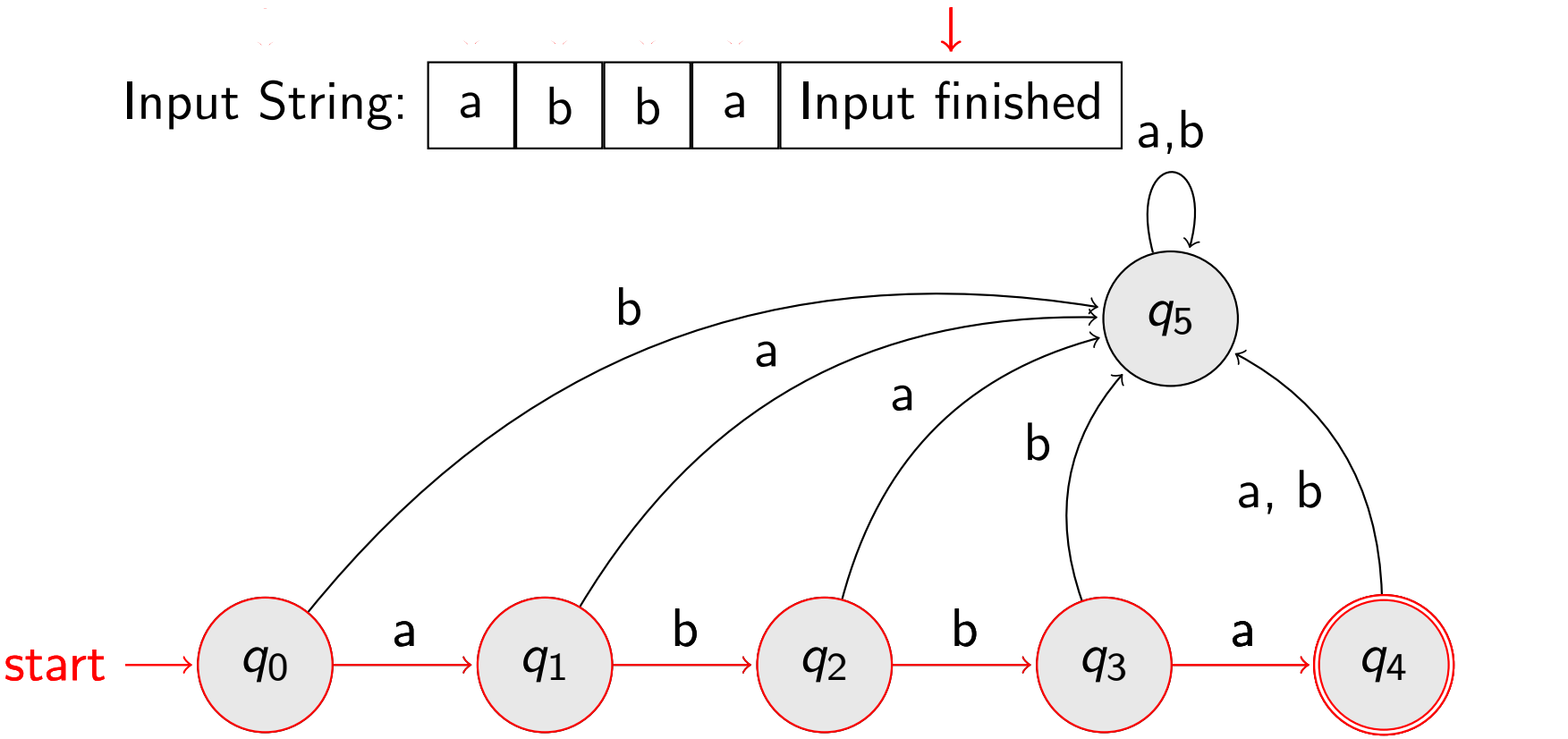

Output: Accept

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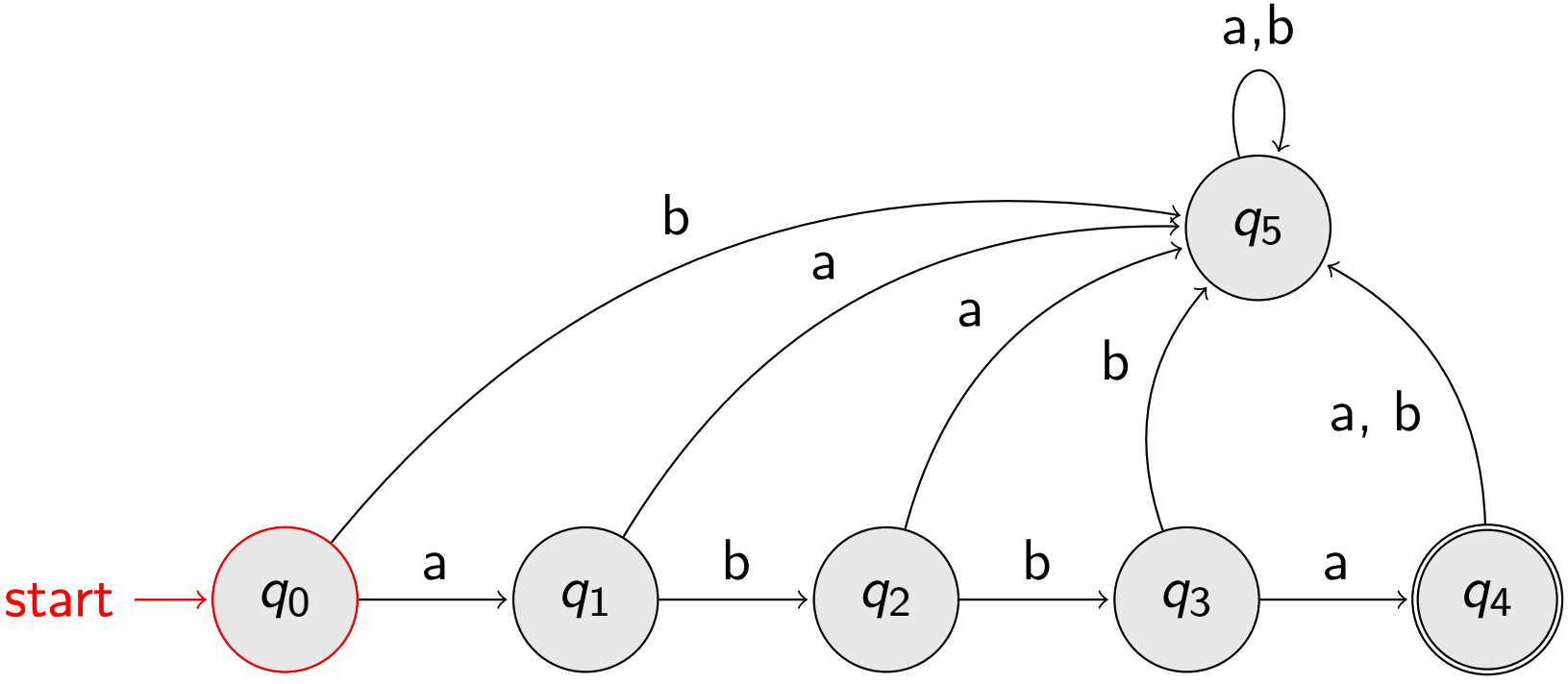
Transition Graph

- **aba** - Reject:

↓

Input String: | a | b | a | Input finished |



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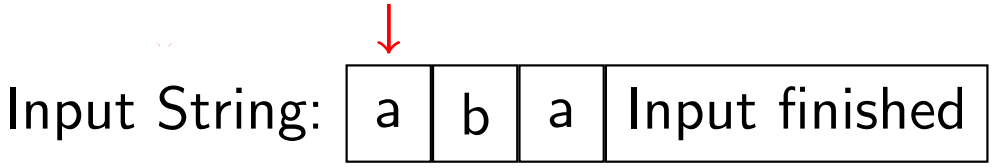

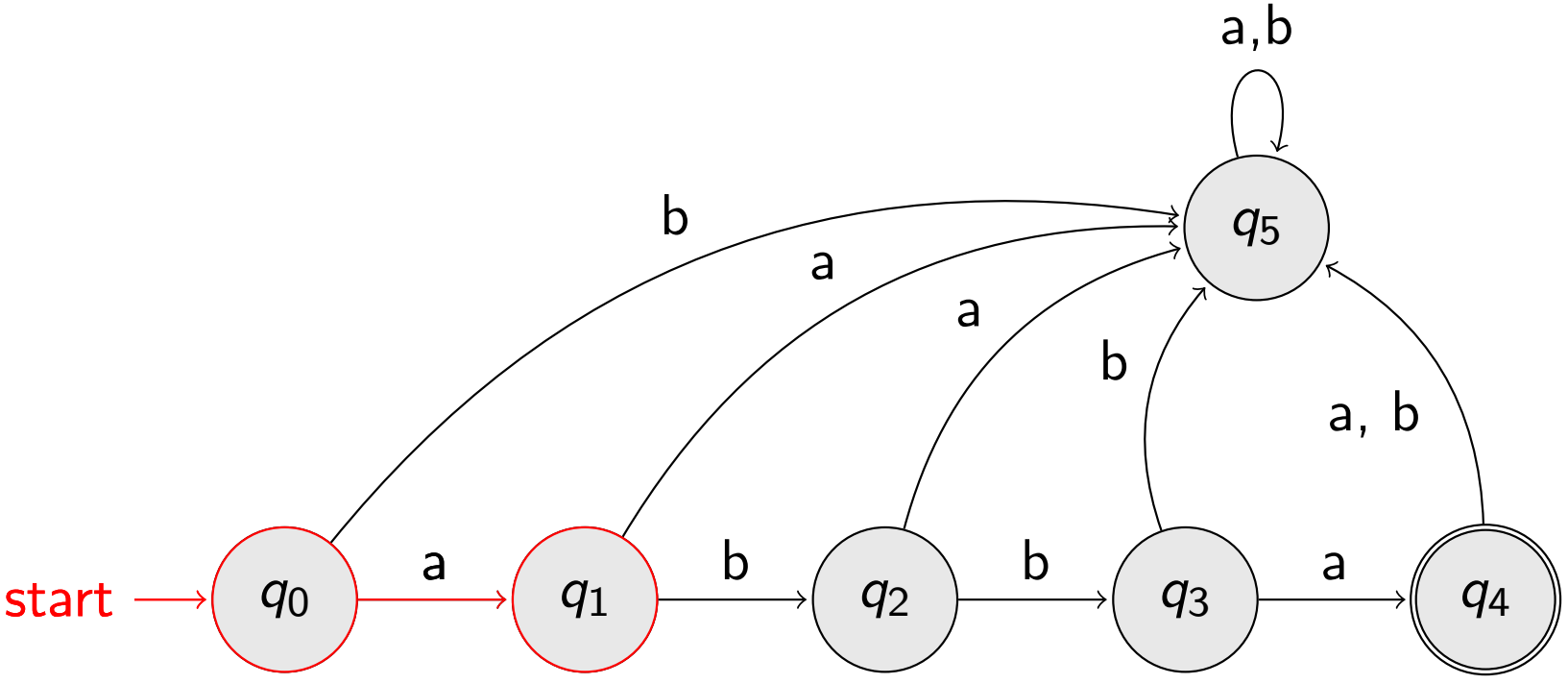

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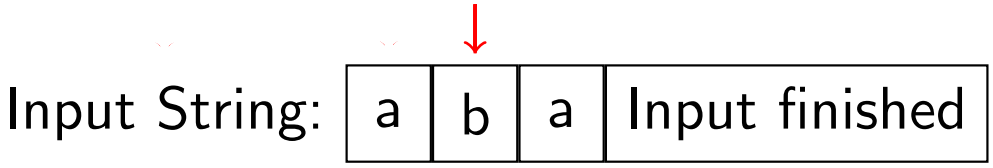
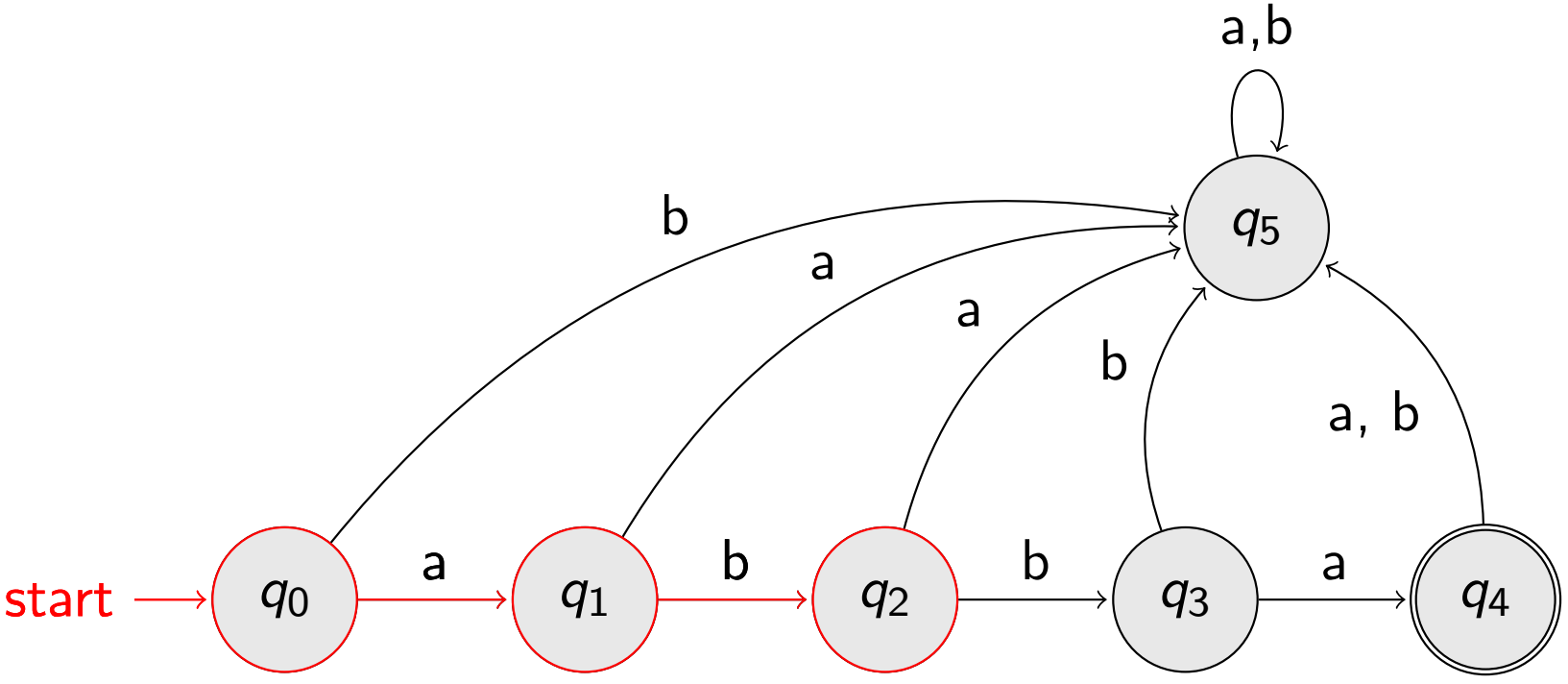
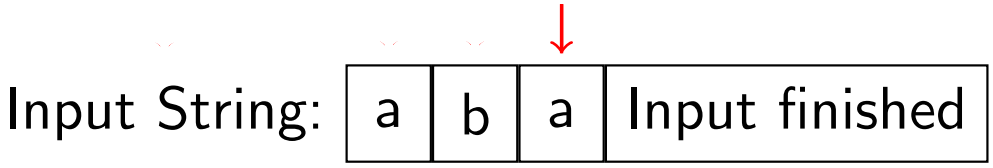
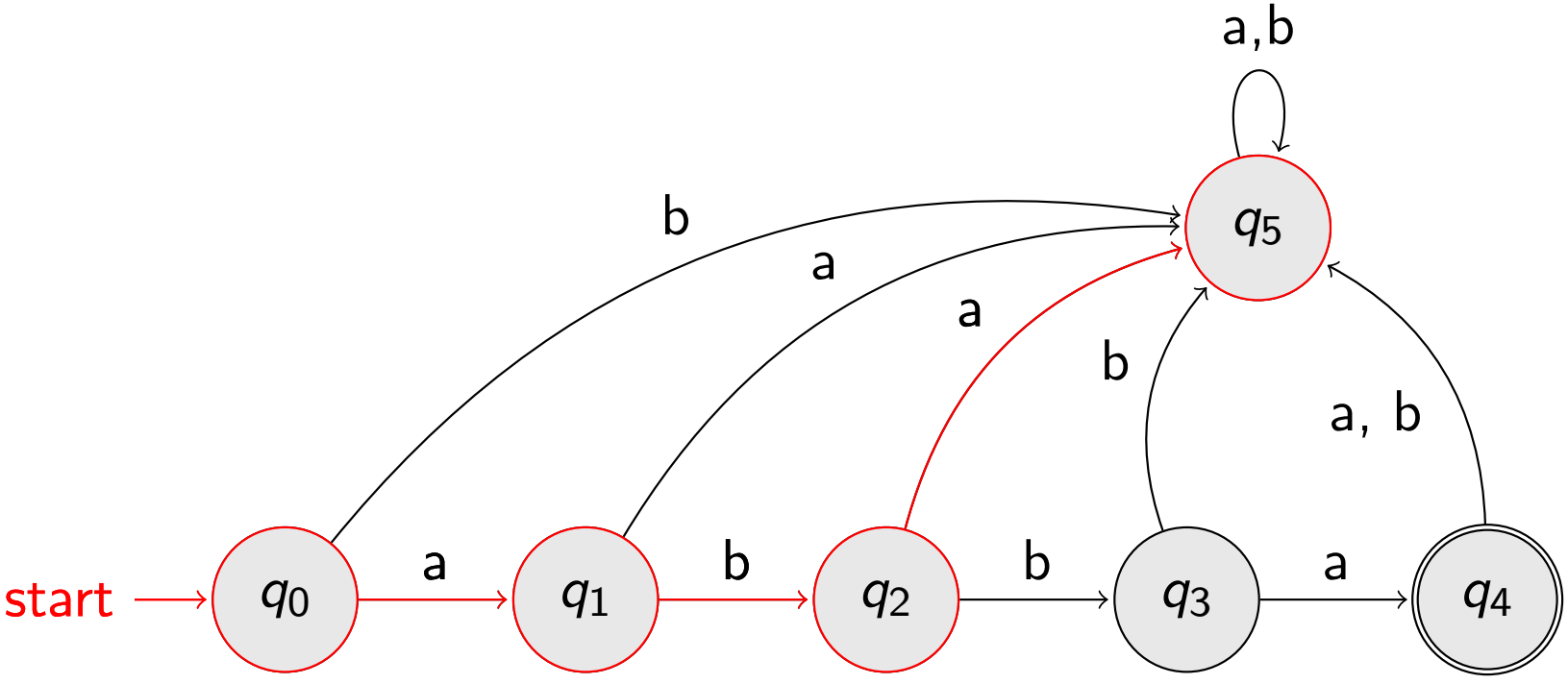




Transition Graph

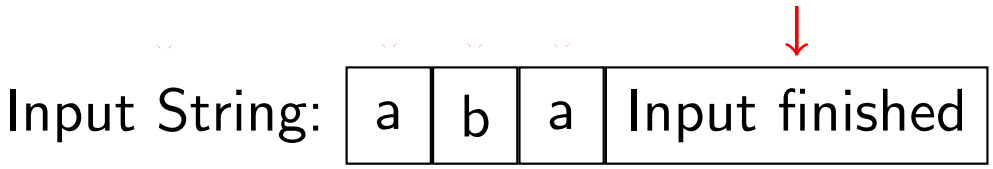
- **aba** - Reject:

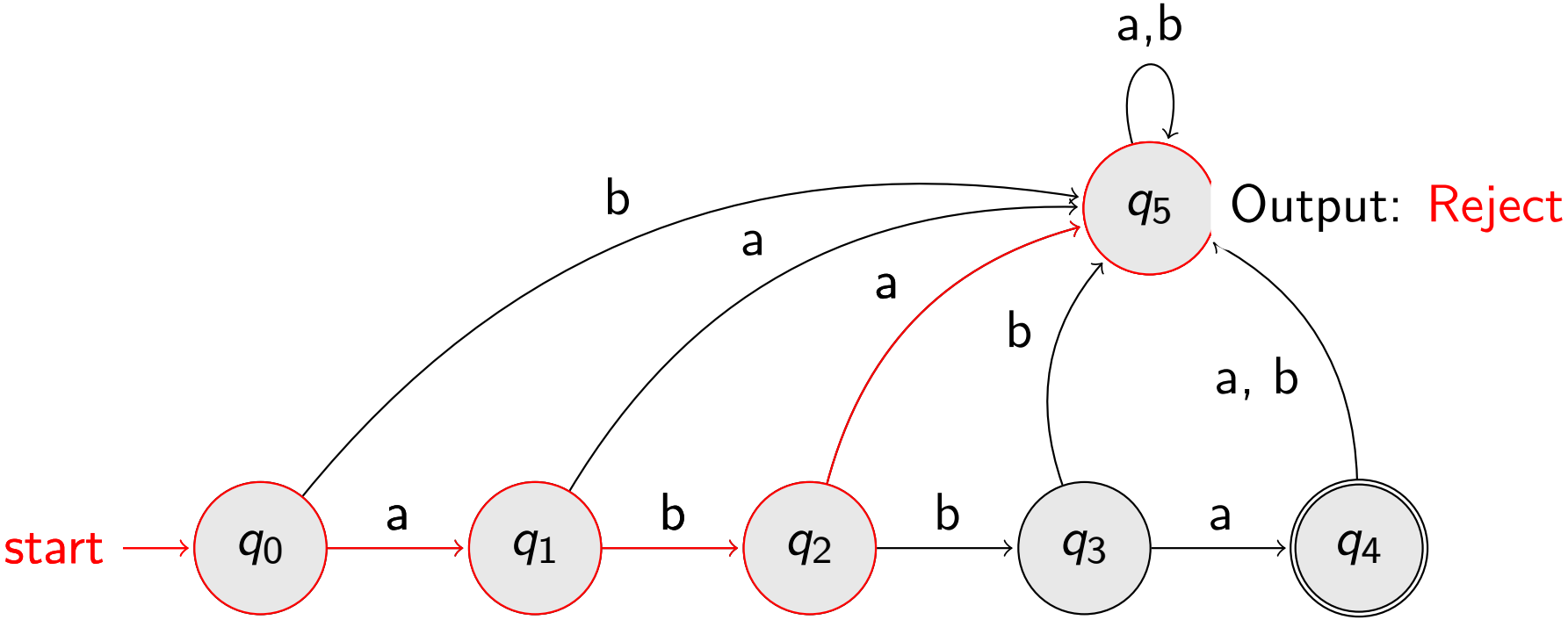




Transition Graph

- **aba** - Reject:





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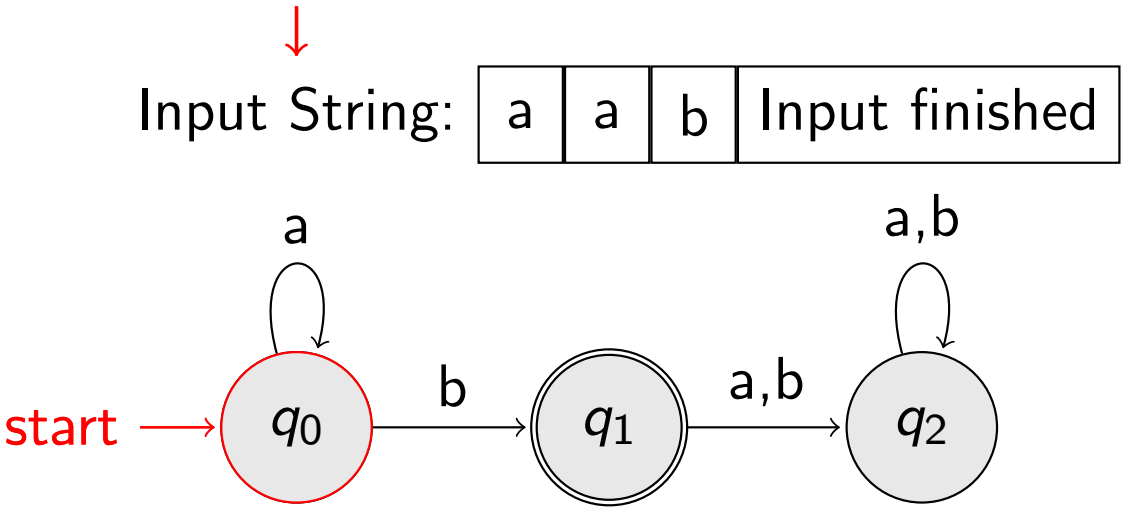
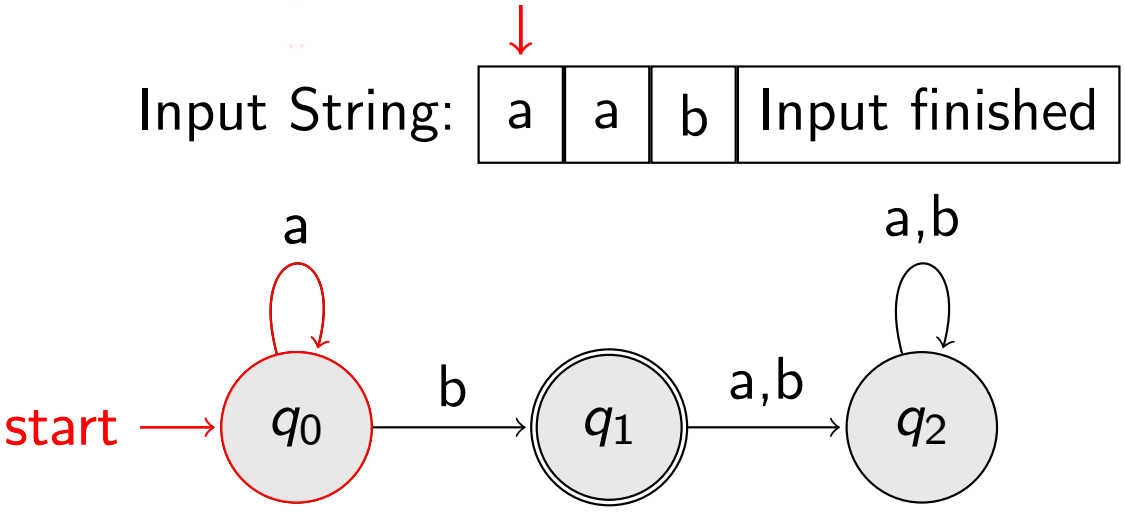
Transition Graph

- **Another Example: Accept**



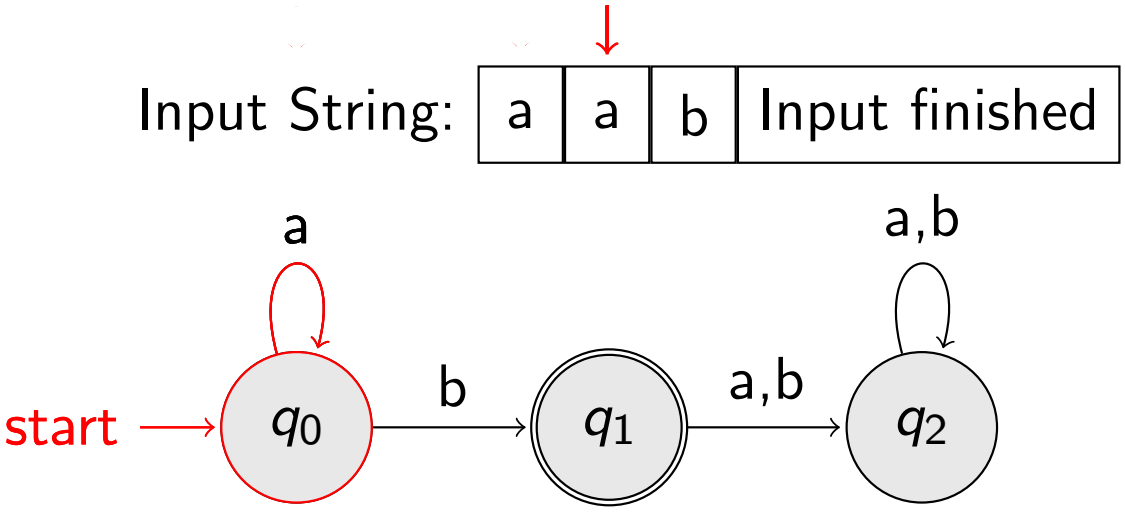
Transition Graph

- **Another Example: Accept**



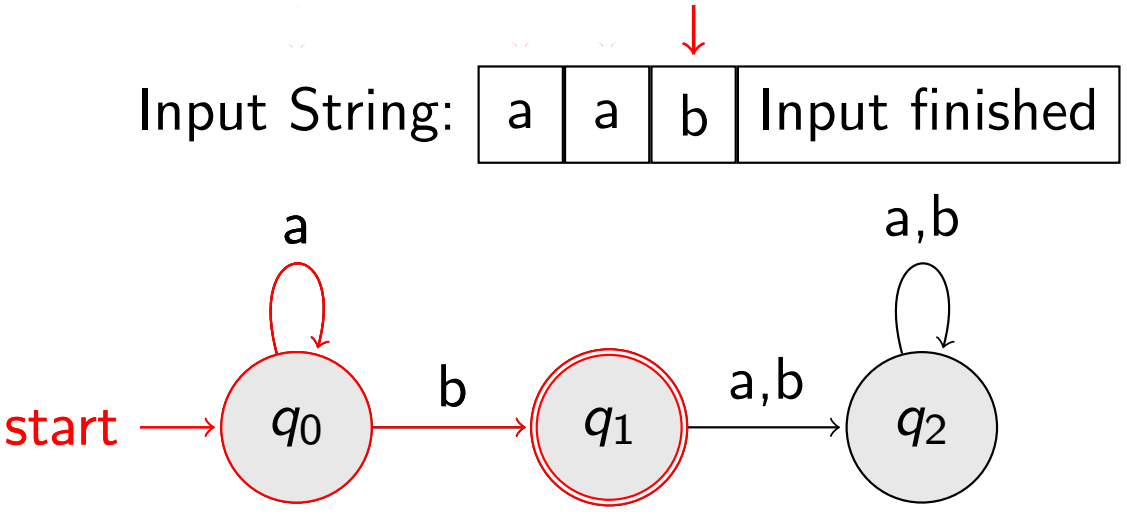
Transition Graph

- **Another Example: Accept**



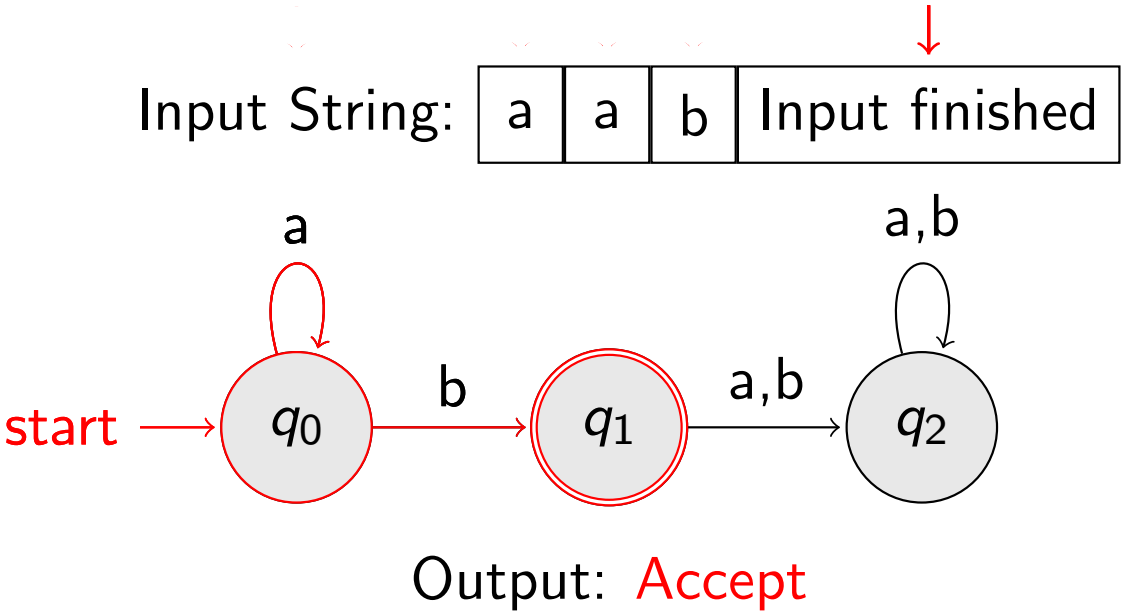
Transition Graph

- **Another Example: Accept**



Transition Graph

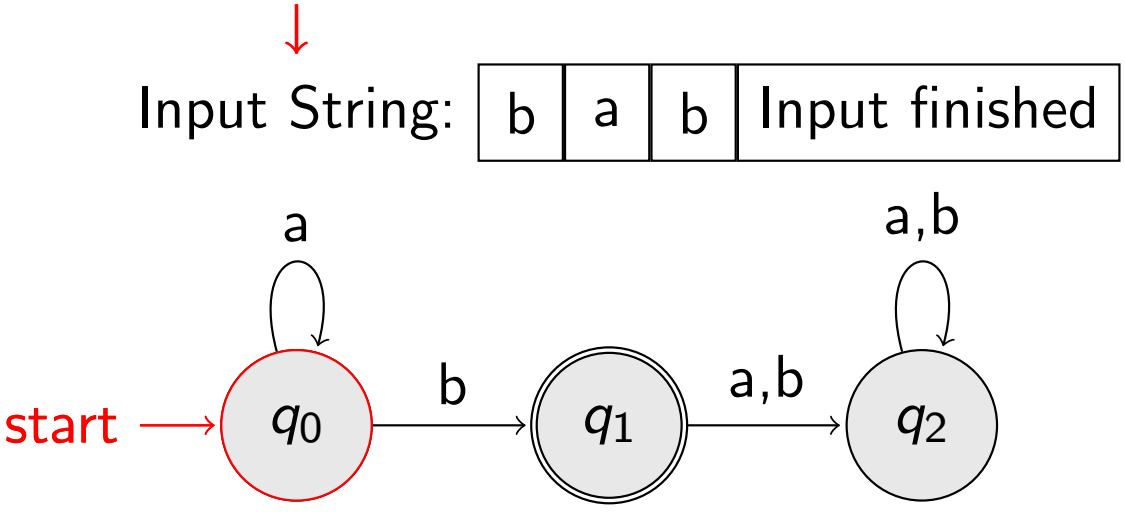
- **Another Example: Accept**



Output: Accept

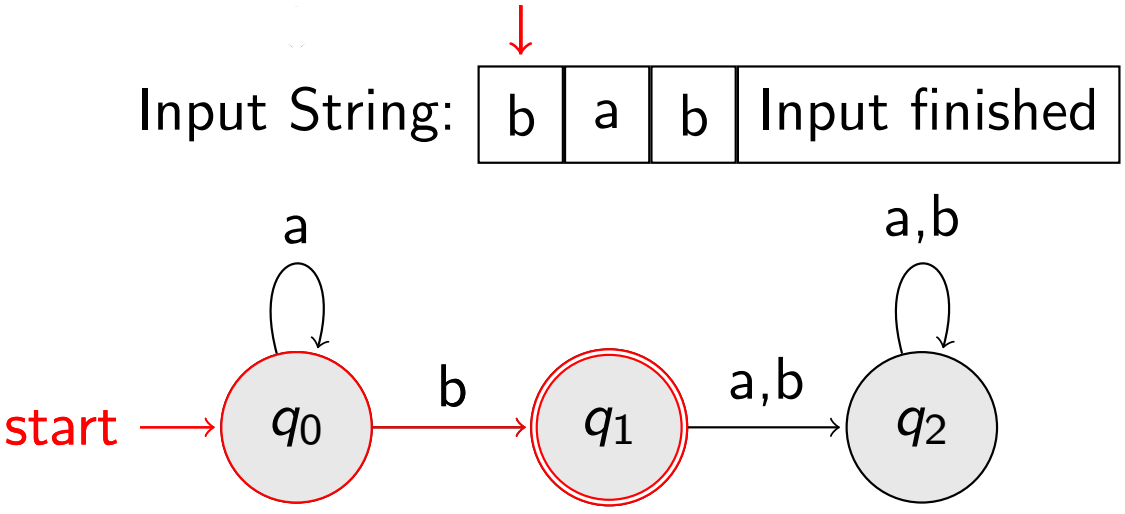
Transition Graph

- **Another Example: Rejection**



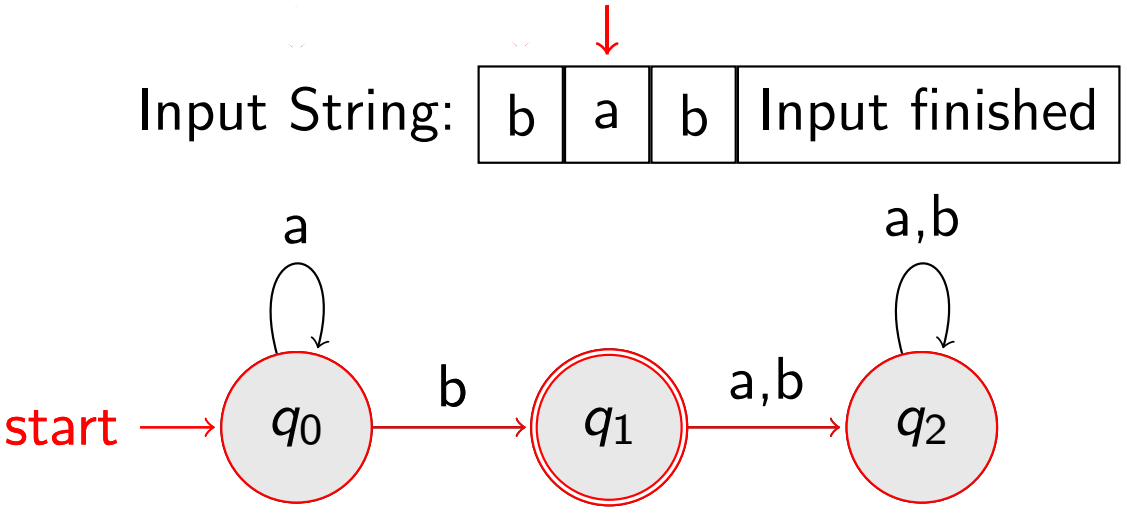
Transition Graph

- **Another Example: Rejection**



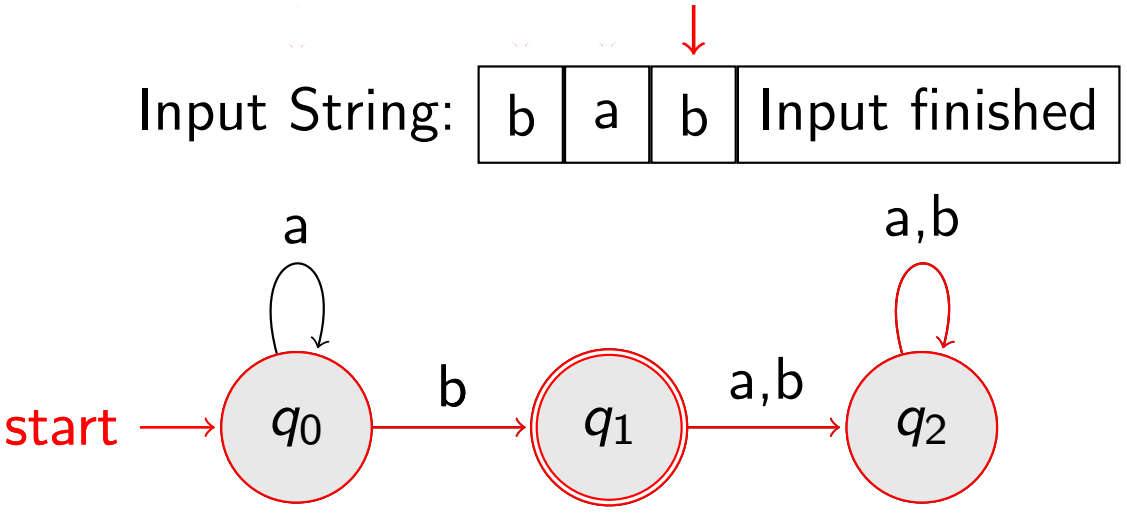
Transition Graph

- **Another Example: Rejection**



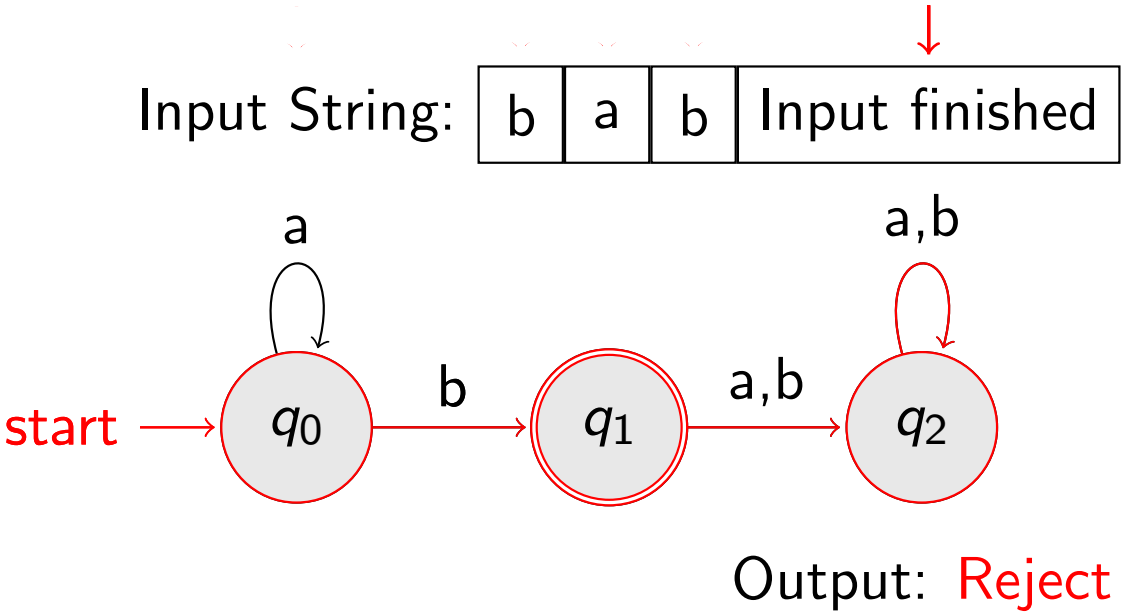
Transition Graph

- **Another Example: Rejection**



Transition Graph

- **Another Example: Rejection**



Output: Reject

Formalities

- **Deterministic Finite Automata (DFA):**

$$M = (Q, \Sigma, \delta, q_0, F)$$

Q: Finite set of states

Σ: Input alphabet

δ: Transition Function: $\delta : Q \times \Sigma \rightarrow Q$

q_0: Initial State

F: Finite set of final states: $F \subseteq Q$

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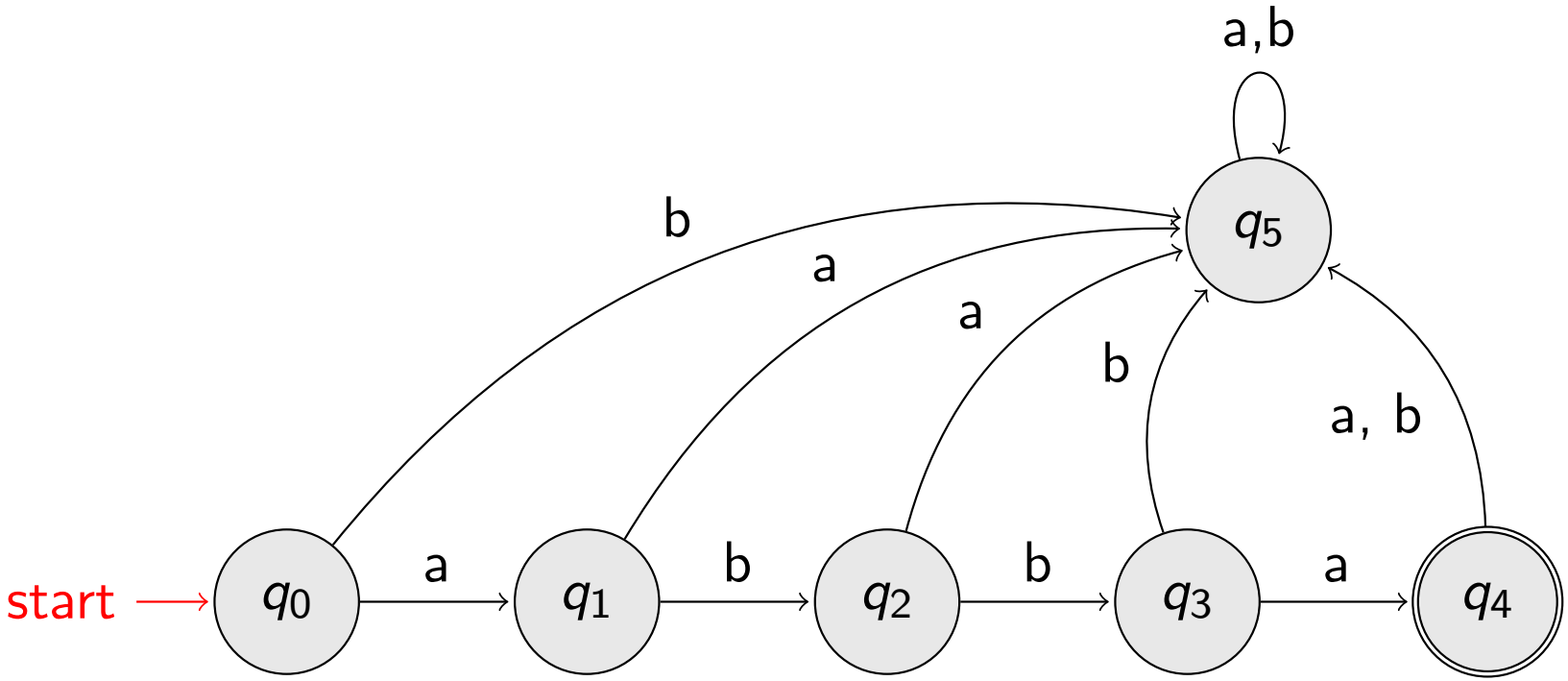
Formalities

- **Finite set of states, Q:**

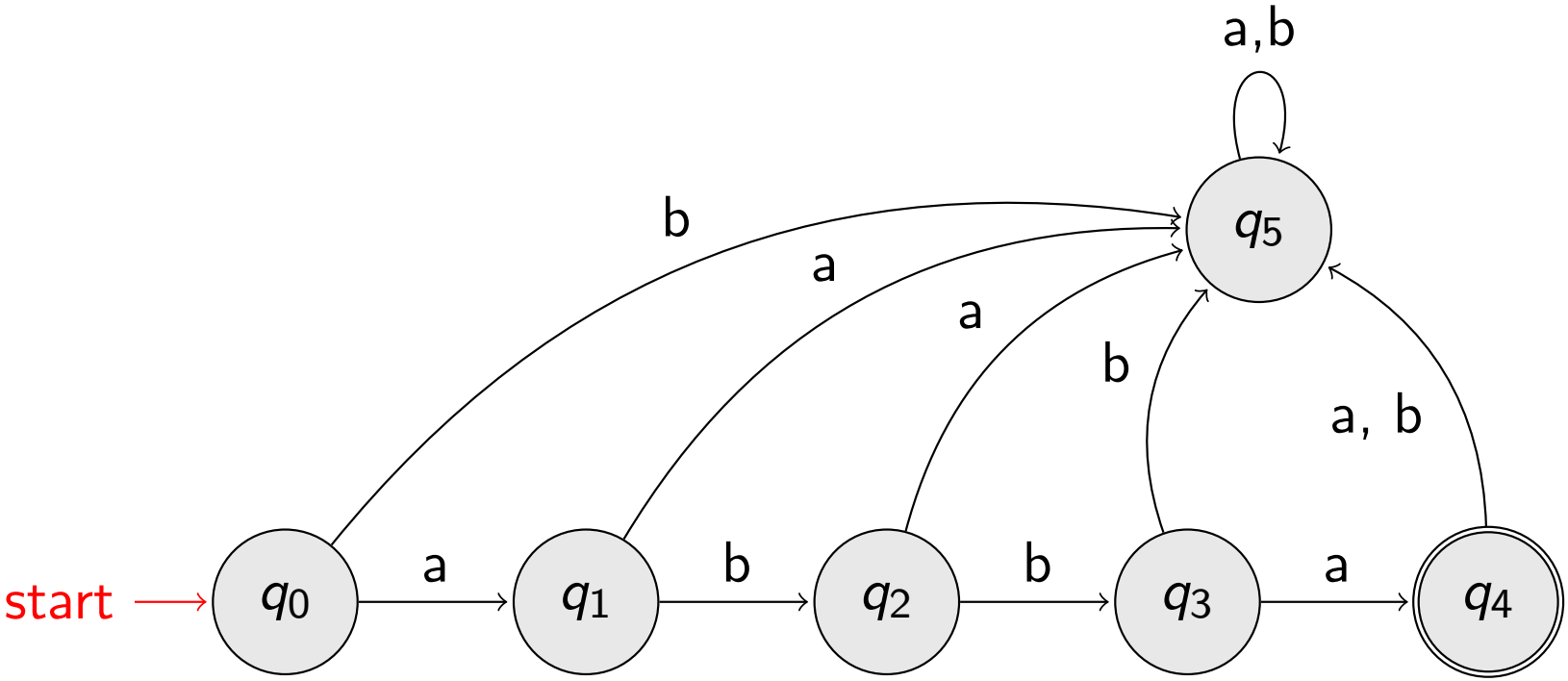
$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$



Formalities

- **Input Alphabet, Σ:**

$$\Sigma = \{a, b\}$$



- **Initial State, $q_0 \subseteq Q$**
- **Finite Set of Final States, F: $F = \{q_4\} \subseteq Q$**

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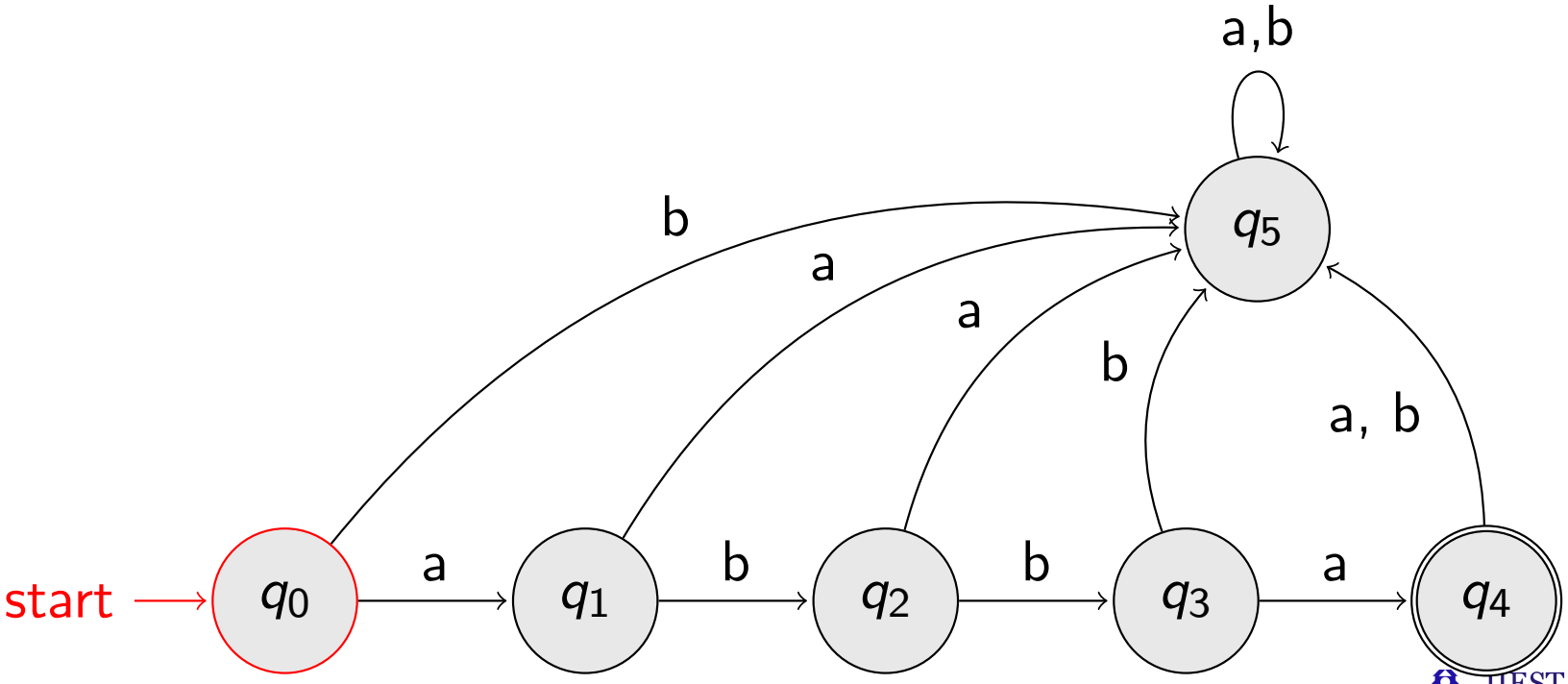
Formalities

- **Transition Function , δ:**

$$\delta : Q \times \Sigma \rightarrow Q$$

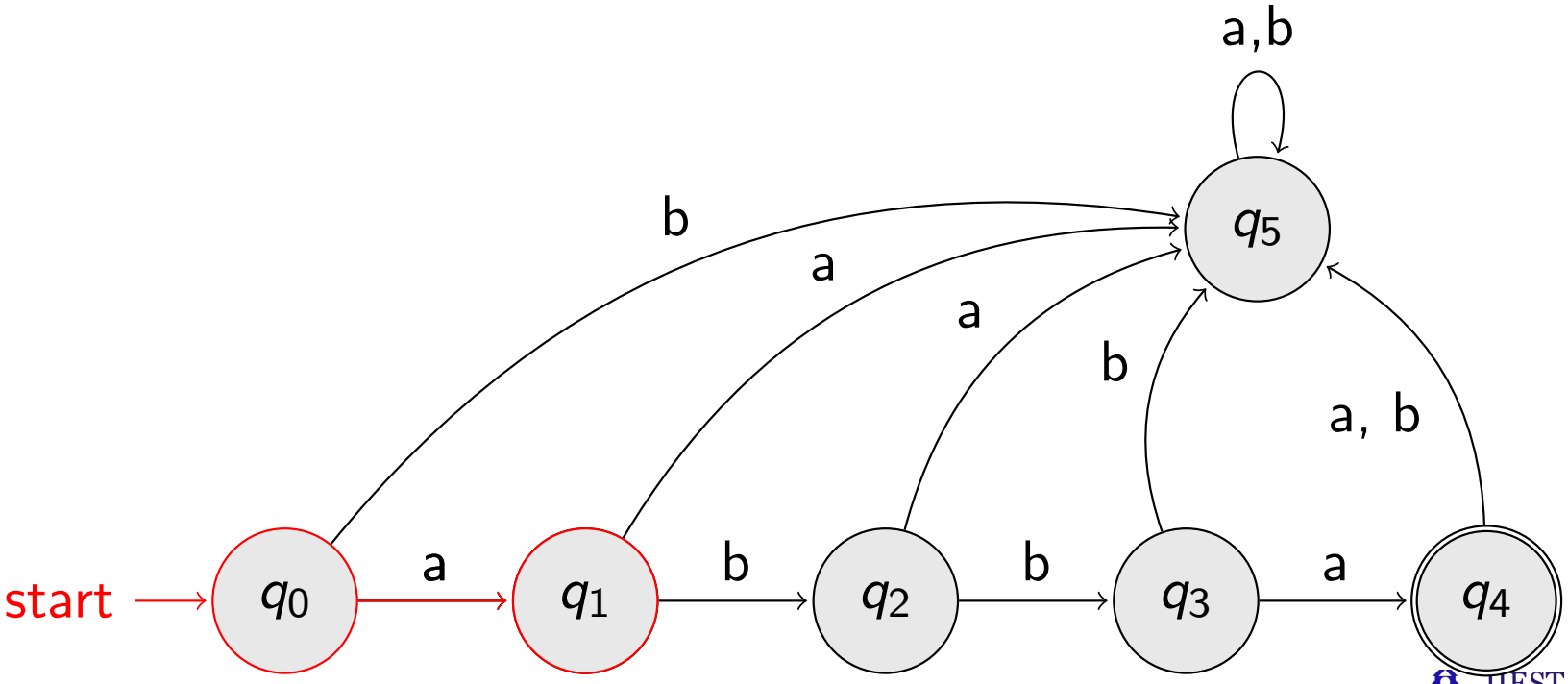


Formalities

- **Transition Function , δ:**

$$\boxed{\delta : Q \times \Sigma \rightarrow Q}$$

$$\delta(q_0, a) = q_1$$



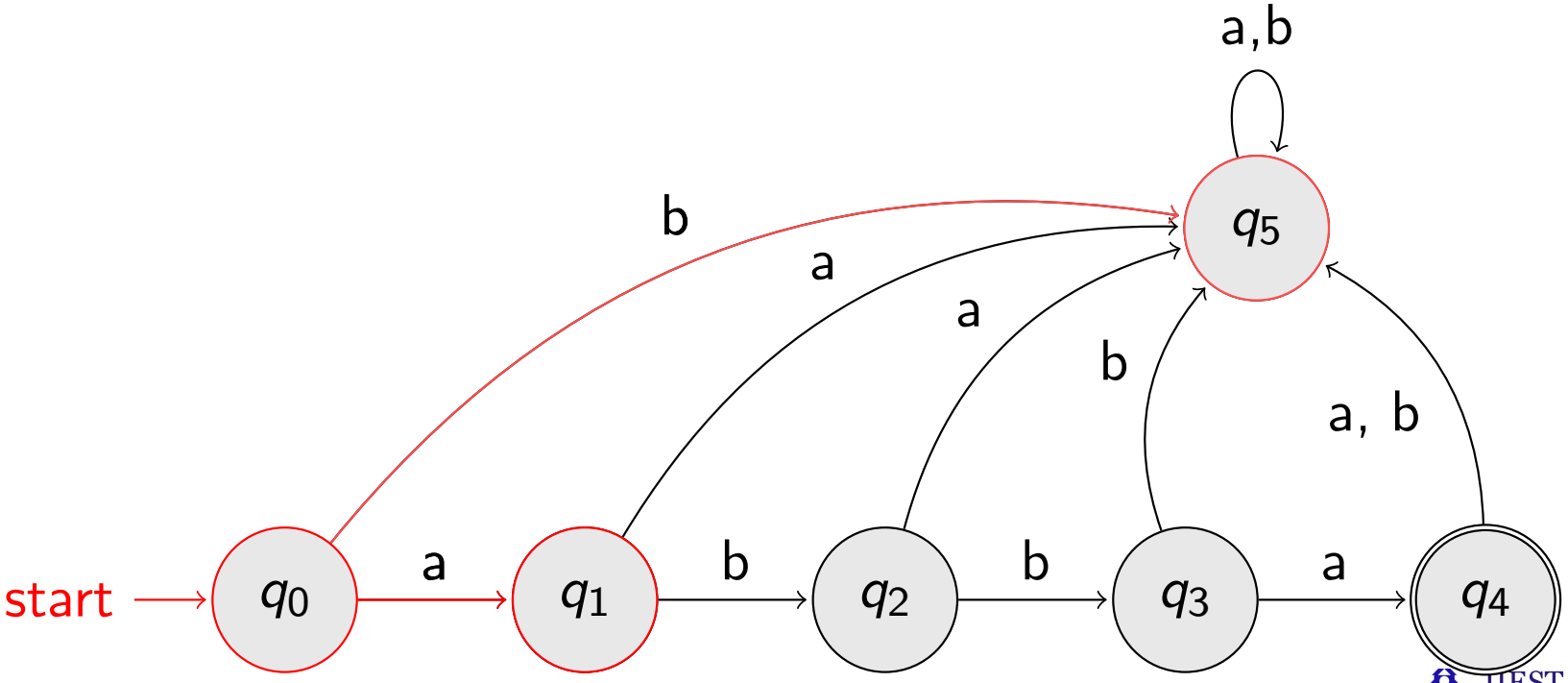
Formalities

- **Transition Function , δ:**

$$\boxed{\delta : Q \times \Sigma \rightarrow Q}$$

$$\delta(q_0, a) = q_1$$

$$\delta(q_0, b) = q_5$$



Formalities

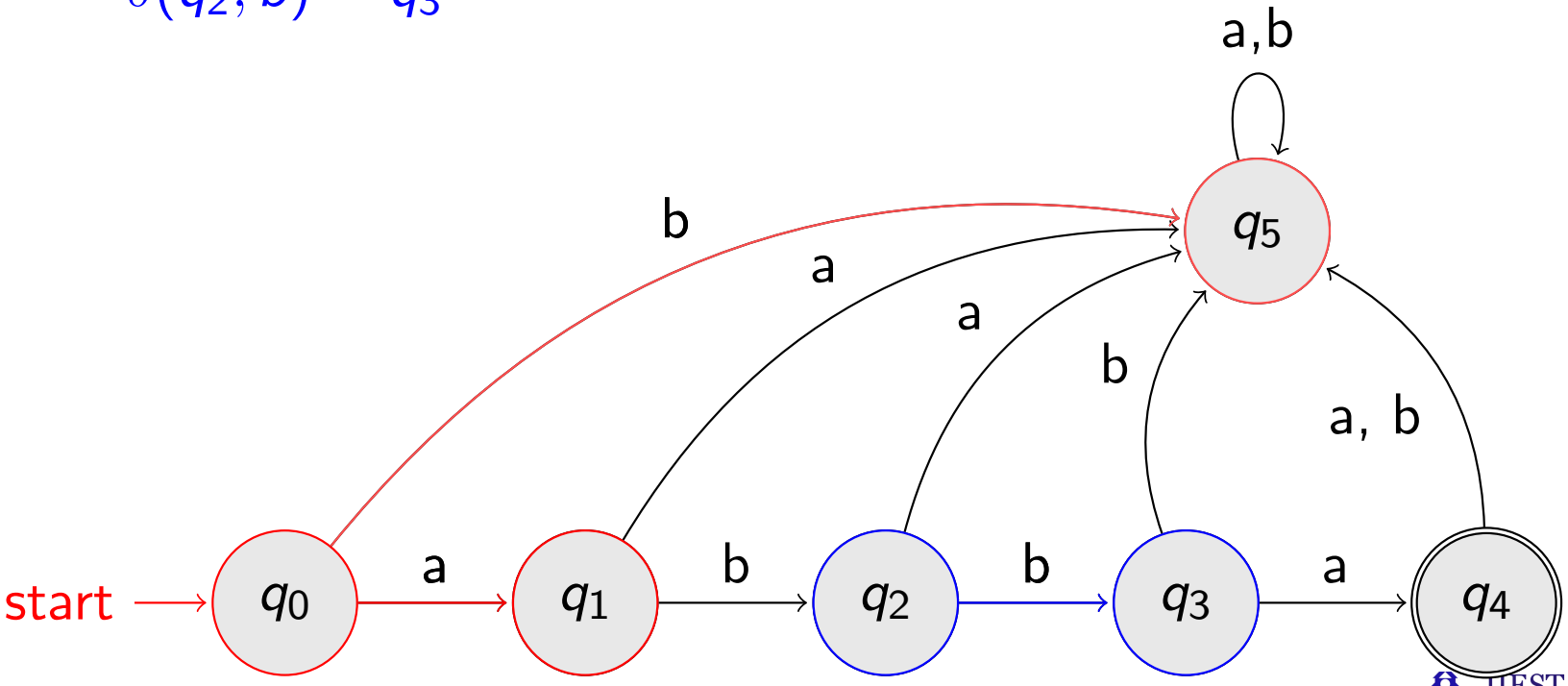
- **Transition Function , δ:**

$$\boxed{\delta : Q \times \Sigma \rightarrow Q}$$

$$\delta(q_0, a) = q_1$$

$$\delta(q_0, b) = q_5$$

$$\delta(q_2, b) = q_3$$

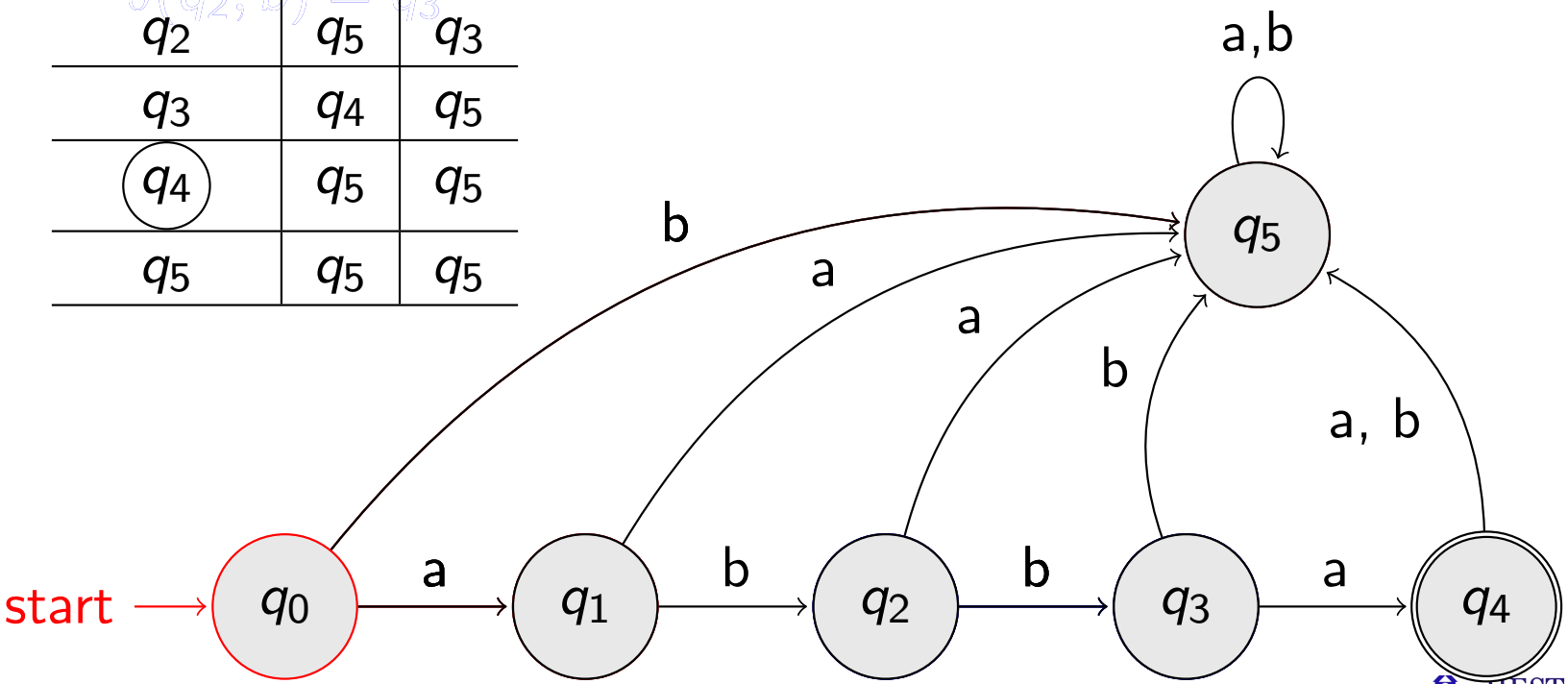


Formalities

- **Transition Function , δ:**

$$\delta : Q \times \Sigma \rightarrow Q$$

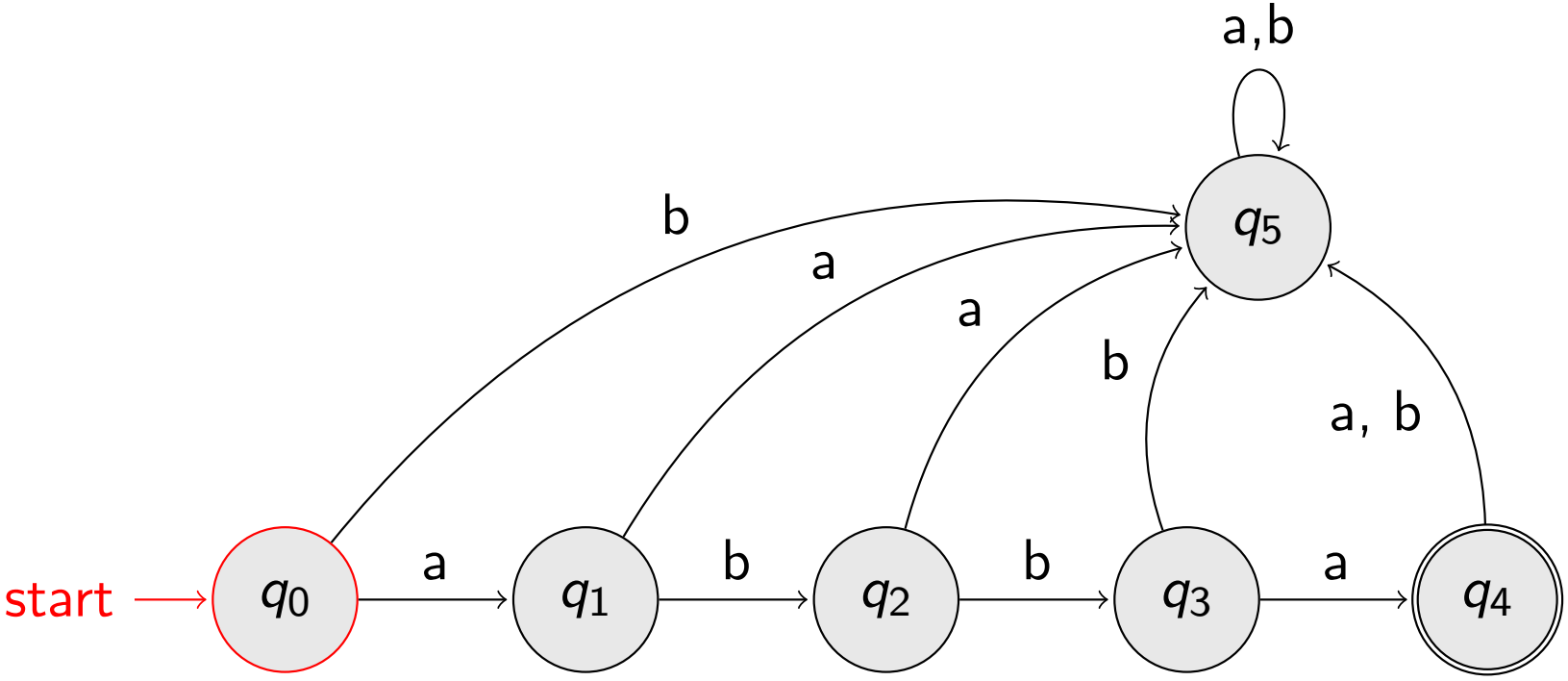
States	a	b
$\rightarrow q_0$	q_1	q_5
q_1	q_5	q_2
q_2	q_5	q_3
q_3	q_4	q_5
(q_4)	q_5	q_5
q_5	q_5	q_5



Formalities

- **Extended Transition Function , δ^*:**

$$\delta^* : Q \times \Sigma^* \rightarrow Q$$

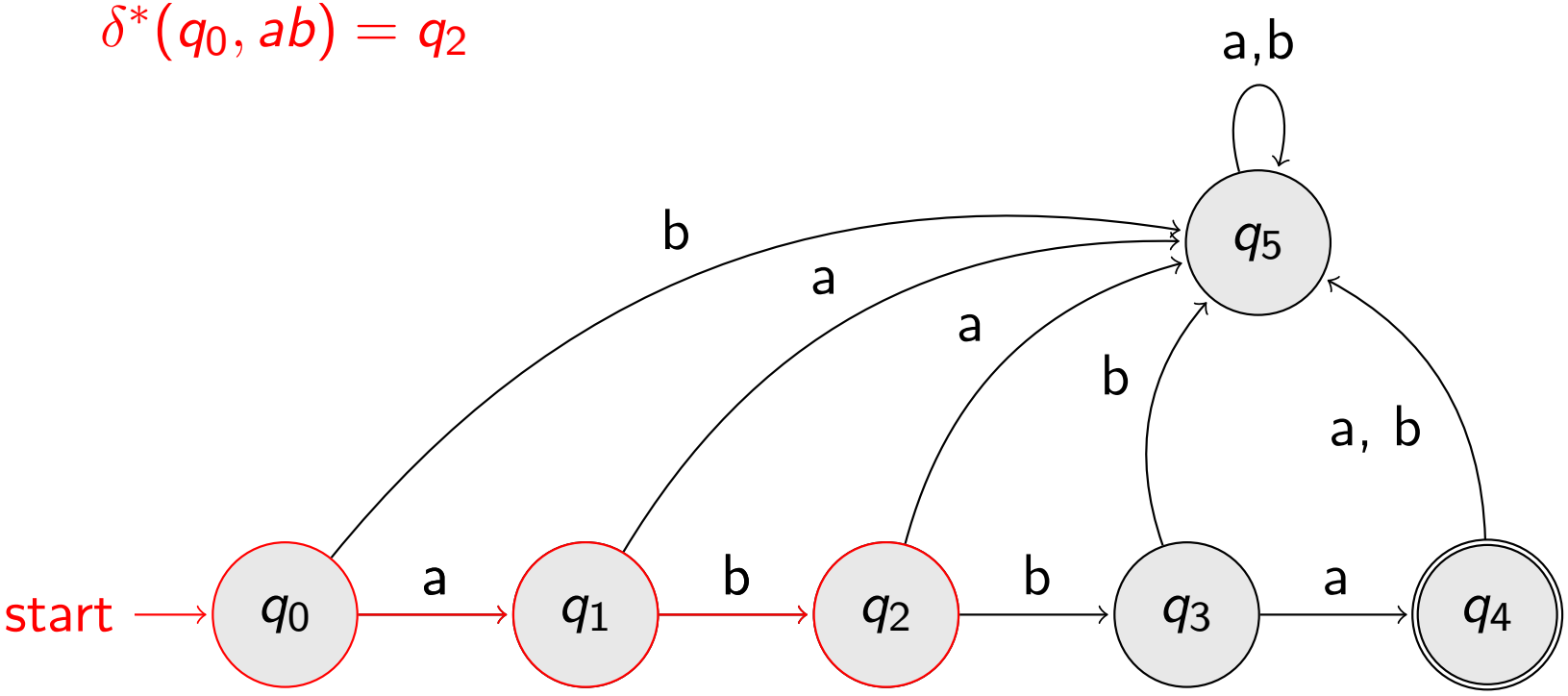


Formalities

- **Extended Transition Function , δ^*:**

$$\boxed{\delta^* : Q \times \Sigma^* \rightarrow Q}$$

$$\delta^*(q_0, ab) = q_2$$



Formalities

- **Extended Transition Function , δ^*:**

$$\boxed{\delta^* : Q \times \Sigma^* \rightarrow Q}$$

$$\delta^*(q_0, ab) = q_2$$

$$\delta^*(q_0, abba) = q_4$$

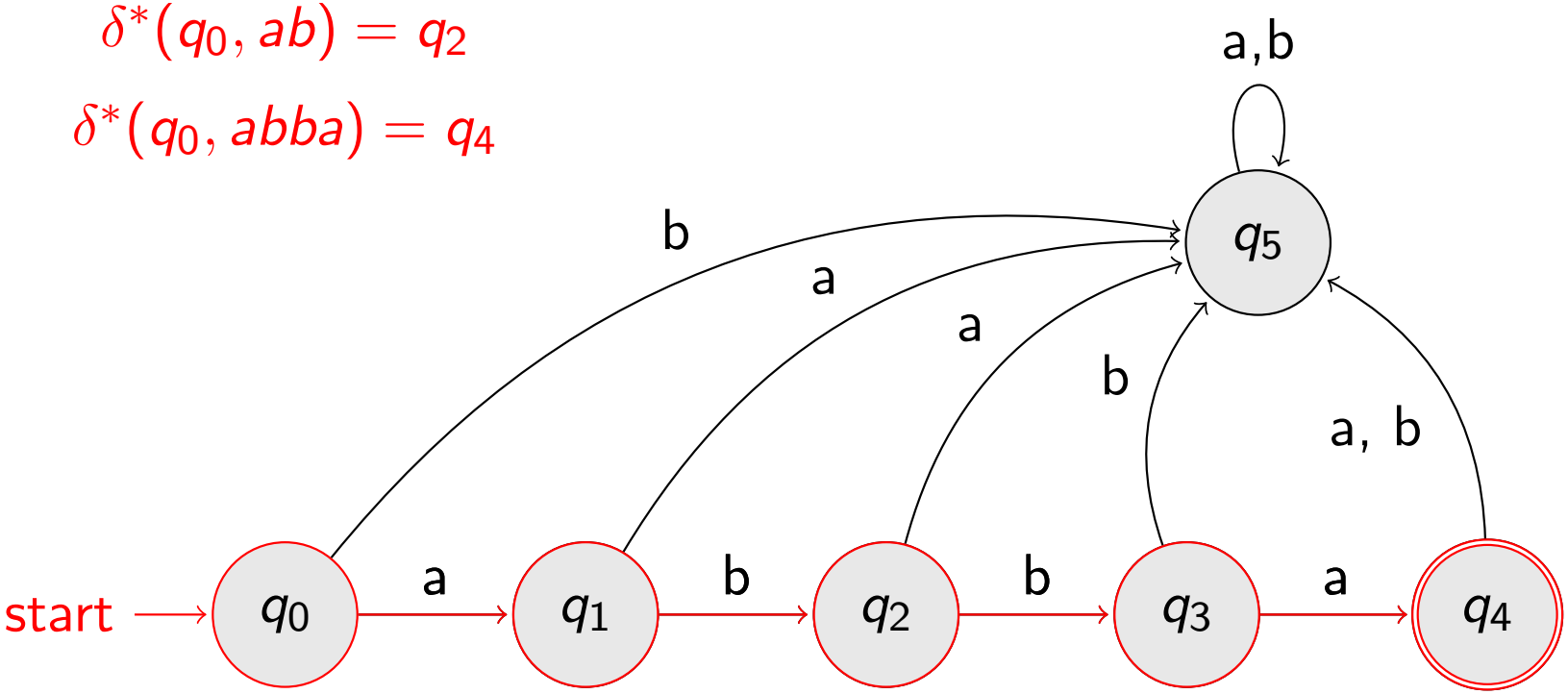


Table of Contents

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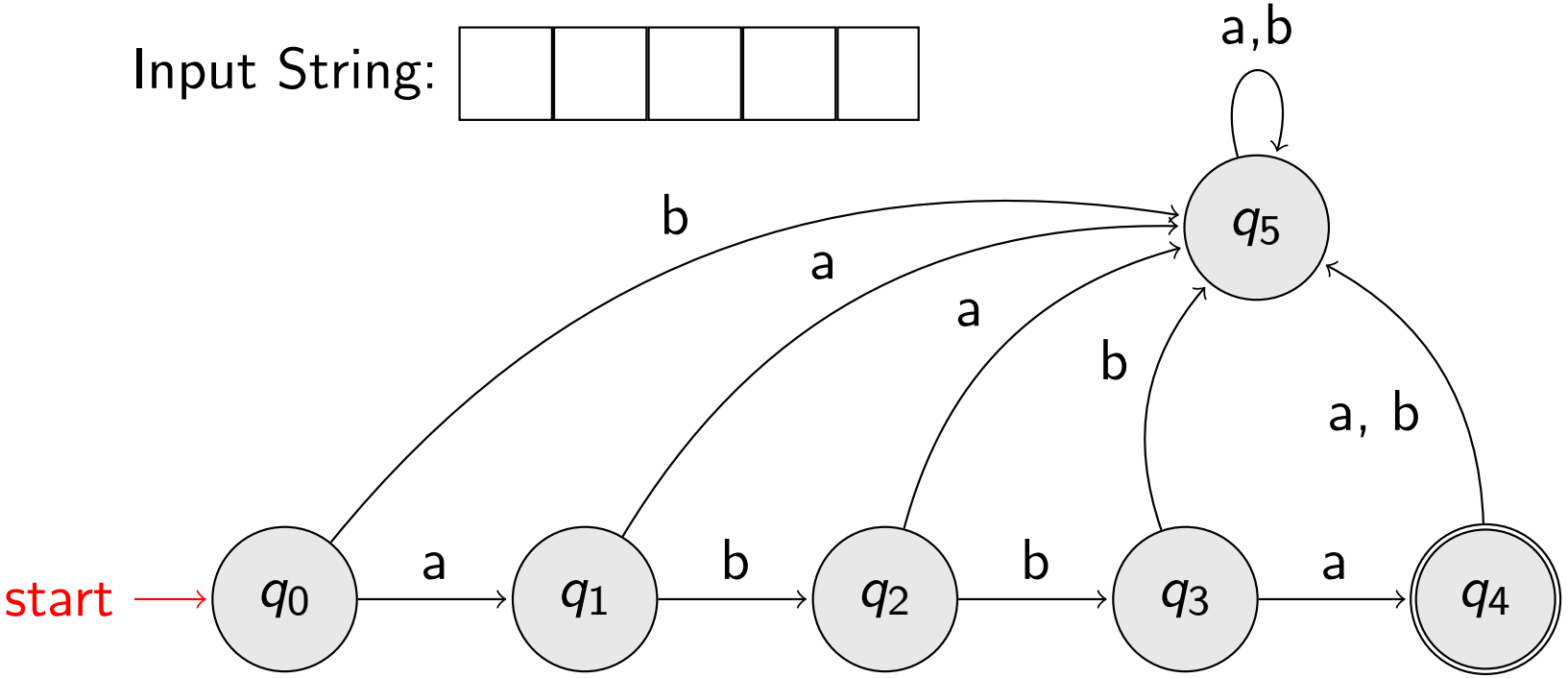
Languages Accepted by DFAs

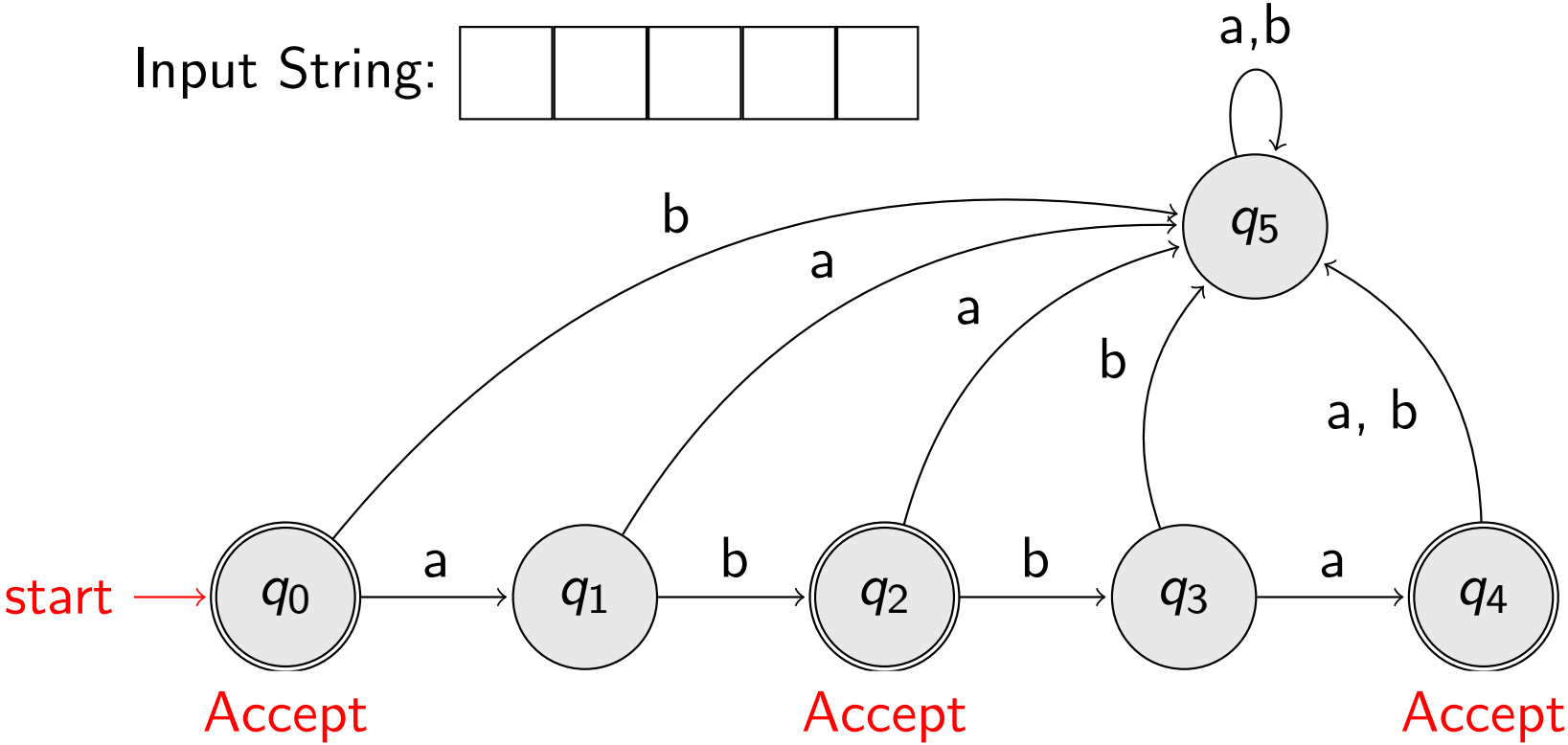
- **Consider a DFA,** M
  - The language, $L(M)$ contains all input strings accepted by M.
  - $L(M) = \{$ strings that drive M to Final State$\}$
- **Example 1:**
  $L(M) = \{abba\}$



Accept

Languages Accepted by DFAs

- **Example 2:** $L(M) = \{\lambda, ab, abba\}$



Languages Accepted by DFAs

- The language accepted by a DFA, $M = (Q, \Sigma, \delta, q_0, F)$ can be defined as follows:
  - $L(M) = \{w \in \Sigma^* : \delta^*(q_0, w) \in F\}$
    Here,
    Q: Finite set of states
    Σ: Input alphabet
    δ: Transition function: $\delta^* : Q \times \Sigma^* \to Q$
    q_0: Initial State
    F: Finite set of final states

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Languages Accepted by DFAs

- **Observations:**
  - Language accepted by a DFA, $M = (Q, \Sigma, \delta, q_0, F)$: $L(M) = \{w \in \Sigma^* : \delta^*(q_0, w) \in F\}$
  - Language rejected by a DFA, $M = (Q, \Sigma, \delta, q_0, F)$: $\overline{L(M)} = \{w \in \Sigma^* : \delta^*(q_0, w) \notin F\}$

References

- Denial I. A. Cohen Introduction to Computer Theory, Second Edition, John Wiley & Sons, 1997.
- J. E. Hopcroft, R. Motwani, & J. D. Ullman Introduction to Automata Theory, Languages, and Computation, Third Edition, Pearson, 2008.